Technical reproducibility report for Clustered hand-hygiene lapses and decontamination efficacy in calibrated ICU transmission models

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Overview

Sections T1–T3 define the synthetic ICU model, hygiene laws and proofs. Sections T4–T10 report coefficient calculations, calibration and robustness studies; T11 covers synthetic measurement uncertainty, T12 stability, T13 reproducibility and T14 directional transfer and worker heterogeneity. All intervals are conditional on the fixed model. They exclude clinical-input and structural uncertainty and, unless stated, calibration-selection and interpolation uncertainty. These simulations are not clinical evidence. "Prespecified" means that a protocol was saved before that study's outcomes; prior studies were often known. Post hoc analyses are identified where reported.

Table T1: Supplement contents. Section links lead to methods, proofs, results and limitations.

T1 ICU model	T8 Route labels
T2 Hygiene laws and opportunity accounting	T9 Environmental comparisons
T3 Proofs	T10 Opportunity-rule studies
T4 Coefficient calculations	T11 Measurement uncertainty
T5 Calibration and matching checks	T12 Stability and platforms
T6 Common-baseline interpolation	T13 Reproducibility
T7 Alternative workflow	T14 Directional transfer and worker heterogeneity

T1 ICU model details and parameter table

T1.1 Structure

The model extends an IDWeek 2022 ICU abstract (Baek et al., 2022) using clinician-informed schedules recovered from that project (Table T2). Transmission passes through workers' hands and bedside sites, never directly between patients. Imported and acquired carriers can transmit. The outcome is new colonization; infection, progression and death are not represented.

One probability p governs all pickup, transfer and site exchange. Contact probabilities are pc and pd, with location weights $c, d \in \{1, .6\}$. The .6 multiplier applies to every patient and site in an isolation location, including susceptible patients. It represents a single-room effect rather than carrier-specific precautions. It has no recovered clinical source and does not mean an observed 40% reduction in acquisition. These are synthetic assumptions.

Table T2: ICU settings common to all simulations unless a section states otherwise. Length of stay is drawn using carriage at admission; a later acquisition does not resample or extend it.

Setting	Declared value
Capacity and staff	30 beds (16 isolation, 14 shared), 30 bedside sites; 10 nurse, 6 physician and 3 ancillary role slots
Model time	2,592 ticks per day (one tick about 33 seconds); fixed activation order
Admissions	One admission draw per empty bed per tick, probability .003; carriage at admission .03
Length of stay	Positive whole-day geometric distribution: mean 5 days for noncarriers and 14 days for carriers at admission
Acquisition and screening	No added stay after acquisition; ideal immediate carrier detection on contact
Isolation	Relocation of detected carriers to isolation beds (rules in Table T3); multiplier .6 for isolation patients and sites
Environment	Binary bedside sites; complete synchronized daily cleaning; no natural decay; initially clean; additional clearing only in the alternative workflow
Hands and hygiene	Initially clean hands; staff-role resets clear hands and keep completion memory; care-zone entry/exit opportunity rule (the inherited per-contact rule only in the clock studies of Section T10); completion law of Section T2
Transmission	Common pickup/transfer/exchange probability p, selected from baseline acquisitions only
Horizon	90 warmup days and 56 observation days for calibration runs, paired control/policy branches and $p = 0$ coefficient backgrounds
Policy	Completion h from .90 (control) to .95 (policy) at day 90, all else fixed

T1.2 Contact process

The contact process was reconstructed from simulator source, following ODD headings (Grimm et al., 2020).

- *Space.* A 32×11 grid with two wings. Each of the 30 beds has an adjacent bedside-site cell

between the bed and its wing's hallway row; a bed and its site form one care zone, fixed to the location rather than the patient. Isolation locations are the 16 beds (and sites) in the outer columns.

- *Agents and scheduling.* 19 workers act once per tick (about 33 seconds) in a fixed activation order: three radiography agents, six physician agents, then ten nurses; beds, patients and sites follow. Workers move one orthogonal cell per tick, first to their hallway row, then along it, then to the destination.
- *Nurses.* The 30 beds are split into ten fixed groups of three (five per wing), one per nurse; each nurse cycles through its three beds without end. During hours 1–15 of each day, at the start of each hour, three designated nurses in each wing are called to the bed whose number equals the hour (repositioning); they resume their rounds once all three have arrived.
- *Physicians.* Six agents are active from midnight for 24, 12, 12, 3, 1 and 1 hours per day; each cycles through a fixed random ordering of all 30 beds, drawn once per run, while active, and returns to its station otherwise.
- *Radiography.* Three agents make rounds through random sets of 30, 20 and 3 beds, starting 2, 1 and 1 times per day; each round ends after its last bed.
- *Contacts.* On each tick a worker contacts every patient and site in its current cell that it has not contacted since entering that cell. Approaching a bed, a worker passes through the zone's site cell, so each visit contacts the site, the patient and the site again on leaving. In the alternative-workflow control arms this gave 103.6 zone visits, 101.1 patient contacts and 207.1 site contacts per occupied patient-day (Table T33).
- *Hygiene.* An opportunity occurs whenever the worker's care zone changes: exit from the old zone, then entry to the new one (Section T2). Contacts within a zone add none.
- *Resets.* Nurses' hands are cleared at the start of each 8-hour slot, physicians' at the start and end of their daily hours, and radiography staff's at the start and end of each round; the completion state carries over, so an incoming slot inherits the outgoing one's state.
- *Transmission and detection.* At each contact a contaminated worker colonizes a susceptible patient or clean site with probability p times the location weight, and a contaminated patient or site contaminates a clean worker with the same probability; any contact with a colonized patient flags it as detected (ideal screening).

T1.3 The two workflows

Table T3 defines the original and alternative workflows.

T1.4 Paired design, outcomes and uncertainty

Each paired job copies the complete state after 90 warmup days into .90 and .95 completion branches, then observes both for 56 days at fixed p. The copies include patients, hands, sites, completion memory and random-generator state. Transmission can change screening, isolation, contacts and subsequent random sequences; event histories can diverge, and no variance reduction is claimed. With acquisitions C_j and patient-days D_j pooled over seeds, $R_j = 1000C_j/D_j$ and

Table T3: Workflow rules. Everything not listed is identical in the two workflows. The alternative workflow is one combined synthetic world; it is not a description of clinical practice, and its two rule changes cannot be separated in this design.

Rule	Original workflow	Alternative workflow
Site cleaning	Complete synchronized daily cleaning only; no cleaning at admission, discharge or relocation. Sites stay at their bed positions and keep their contamination when patients move.	Daily cleaning as in the original, plus clearing of a bedside site when its patient leaves by discharge or relocation (contamination and its origin label are cleared; the daily clock is not reset). Moving in or a first admission to a never-occupied bed triggers no cleaning.
Isolation relocation	After each tick, a detected carrier in a shared bed moves to the first isolation bed, in fixed order, that holds no detected carrier. Any occupant there is swapped into the vacated shared bed. If every isolation bed holds a detected carrier, no move occurs.	A detected carrier in a shared bed moves only into an empty isolation bed (first in the same fixed order). If none is empty it stays in its shared bed, with that bed's multiplier; the rule is re-applied after every tick, so it moves once an isolation bed is empty. No occupant is ever displaced.
Consequence for the bed a patient enters	A swapped or newly admitted patient can meet contamination left at the destination bed by a previous occupant.	Contamination left by a previous occupant is cleared, except deposits made while the bed stood empty.

$F = 1 - R_1/R_0$, undefined when $C_0 = 0$. We also report $R_0 - R_1$ and mean paired acquisitions prevented per run, because relative and absolute benefits can rank differently.

All intervals in this supplement describe Monte Carlo sampling in a fixed synthetic model; they exclude clinical-input, structural, calibration-selection (unless stated) and interpolation uncertainty. Whole seed vectors are resampled, keeping paired arms and conditions together. Most intervals are descriptive 95% intervals. Each study fixed its own families in advance: three between-profile F contrasts at 98.333% (original holdout, alternative study and common-baseline analysis), three environmental contrasts at 98.333%, six follow-up contrasts at 99.167%, four local-sensitivity contrasts at 98.75%, six original matching checks at 99.167% and twelve inherited-clock sampled-match checks at 99.5833%. Precision conventions: prevention is reported to one decimal except where the information would be lost (leading-order limits to two decimals and paired between-workflow differences in limits to three).

T2 Hygiene laws and the stationary Markov chain

Completion process. Each worker carries a sequence of hygiene opportunities with completion indicators $X_t \in \{0, 1\}$ (0 = miss, 1 = completion). The sequence is a two-state Markov chain with mean completion h and lag-one autocorrelation ρ:

$$P(h, \rho) = \begin{pmatrix} \rho + (1-\rho)(1-h) & (1-\rho)h \\ (1-\rho)(1-h) & \rho + (1-\rho)h \end{pmatrix}, \tag{T1}$$

rows and columns ordered (miss, completion). After a completion the next completion probability is $h + \rho(1-h)$; after a miss it is $h(1-\rho)$. The stationary law is $\pi = (1-h, h)$. The study

uses $\rho \in [0,1]$; negative values would require $\rho \geq -\min\{h/(1-h), (1-h)/h\}$ for valid transition probabilities. At $\rho = 1$ the chain is reducible and the specified stationary mixture is used. The process advances at every opportunity, even when hands are clean. The first-ever opportunity of a worker uses a Bernoulli(h) state; staff-role resets clear hands but keep the completion state; at the policy switch h changes and the current state is kept.

Clearing. A completed action clears the hand with probability e, independently of everything else given the completion sequence; a miss never clears. With $\Gamma(e) = \mathrm{diag}(1, 1-e)$, survival of a contaminated hand through k subsequent opportunities, started from the completion-state law ν at the preceding opportunity, is

$$q_k(\nu; h, \rho, e) = \nu \left[P(h, \rho)\Gamma(e)\right]^k \mathbf{1}, \qquad q_0 = 1, \tag{T2}$$

and under stationarity $q_k = \pi\Gamma(P\Gamma)^{k-1}\mathbf{1}$ for $k \geq 1$. Thus $e = .8$ means a 20% chance that a completed action leaves the hand contaminated, not an 80% reduction in bacterial load. Special cases: independent completion ($\rho = 0$) gives $q_k = (1-he)^k$; perfect clearing ($e = 1$) gives $q_k = (1-h)[\rho + (1-\rho)(1-h)]^{k-1}$; and in general

$$q_1 = 1 - he, \qquad q_2 = (1-he)^2 + e^2\rho\, h(1-h), \tag{T3}$$

because $q_2 = \mathbb{E}[(1-eX_1)(1-eX_2)]$ and $\mathbb{E}[X_1X_2] = h^2 + \rho h(1-h)$.

Profiles. The primary assumptions are independent misses $(\rho, e) = (0, 1)$, clustered misses $(.5, 1)$ and imperfect clearing $(0, .8)$; none estimates staff behaviour. Under independence survival depends on he, the usual compliance-times-efficacy parameter (Austin et al., 1999; Beggs et al., 2009). Independent and clustered misses instead share h and q_1 but differ from q_2 onward (Table T4).

Table T4: Stationary survival at $h = .90$ and isolated-bridge prevention for completion .90→.95 (exact rational arithmetic; % of survival removed). $F_k = 1 - q_k(.95)/q_k(.90)$. The combined profile is the fourth truth of the measurement study.

Profile	(ρ, e)	q_1	q_2	q_6	F_1 (%)	F_2 (%)	F_6 (%)
Independent misses	(0, 1)	1/10	1/100	10^{-6}	50.0	75.0	98.4
Clustered misses	(.5, 1)	1/10	11/200	0.00503	50.0	52.3	60.4
Imperfect clearing	(0, .8)	7/25	49/625	0.000482	14.3	26.5	60.3
Combined	(.5, .8)	7/25	67/625	0.00840	14.3	32.1	58.7

Opportunity accounting. Model entry corresponds to WHO Moment 1; exit corresponds to Moments 4 and 5. Within-zone Moments 2 and 3 are absent. WHO observation merges coinciding indications into one opportunity, met by either action (WHO, 2009a, Sections II.5.6 and III.5). Exit A and entry B merge only for a direct move without touching surfaces outside the zones (WHO, 2009b, Part I, Section 21.4.2, p. 102). All model moves are direct because no surfaces outside zones are contactable. Real audits mix direct and indirect moves and Moments 2 and 3; neither model score maps to audited compliance without those proportions.

For direct moves, the miss probability is $(1-h)\{\rho + (1-\rho)(1-h)\}$, equal to q_2 at $e = 1$. A transition audit measures this behavioural component; a boundary audit measures h (Table T5). Under independence, $h = 1 - \sqrt{1-c}$ converts transition compliance c to per-boundary completion:

90% and 95% give approximately .684 and .776. At perfect clearing this transition policy halves two-opportunity survival for every ρ, so $F_2 = 50\%$. Its full-ICU comparison, including imperfect clearing, is reported in Section T4.4. Different compliance definitions define different interventions; comparisons across ρ are like-for-like only within one definition. This correspondence assumes a stationary Markov law and one exit and one entry draw per direct move. Transition audits alone identify neither h nor e.

Electronic systems score events differently. Crediting one dispense to an exit and the next entry shifts scores toward transition units; independent event misclassification attenuates estimated ρ. Shared bays may be harder to instrument, and group systems lack worker sequences. Estimate ρ from raw per-worker records, not vendor scores. Consecutive miss runs also occur under independence, with geometric lengths (Chan et al., 2026); runs alone do not show serial dependence. Lag-one autocorrelation measures dependence, and lag two helps check the Markov assumption. Unless specified otherwise, policies here raise per-boundary h with ρ fixed; holding the miss-to-complete probability or mean miss-run length fixed would change the response.

Table T5: Model completion per boundary event and the corresponding WHO-style transition compliance (an exit followed directly by an entry counted as one opportunity, met if either action occurs). Closed forms; the transition miss probability equals q_2 when $e = 1$.

Profile (ρ, e)	Per-boundary h	Transition compliance (%)	Transition miss $= q_2$ $(e = 1)$
Independent misses (0, 1)	0.90	99.00	0.0100
	0.95	99.75	0.0025
Clustered misses (0.5, 1)	0.90	94.50	0.0550
	0.95	97.38	0.0263
Imperfect clearing (0, 0.8)	0.90	99.00	—
	0.95	99.75	—
Combined (0.5, 0.8)	0.90	94.50	—
	0.95	97.38	—

Bridge survival in the ICU. In the full model a bridge is a pickup from a carrier followed by a contact of the same worker with a susceptible patient before a role reset. Its survival is the ordered product $\nu_i \prod_m [P(h_m, \rho)\Gamma(e_m)]\mathbf{1}$ over its intervening opportunities, with the actual preceding completion-state law ν_i; the policy switch can fall inside a bridge. Stationary q_k is an additional approximation, used only as a comparator; the coefficient computation of Section T4 uses the exact law.

T3 Proofs

The results in this section formalize, for a sequenced ICU contact record, quantities that are standard in simpler settings: the per-journey product of pickup, survival and transfer probabilities (Beggs et al., 2009), the host–vector structure of hand-mediated transmission in which hygiene shortens the "vector" contamination period (Austin et al., 1999), and linear-programming and moment bounds from partial identification (Manski, 2003; Tamer, 2010; Bertsimas and Popescu, 2005; Madansky, 1959). No new probability theory is claimed; the contribution is the application. Each statement lists its assumptions.

T3.1 Rare transmission on a fixed contact record

Proposition T1 (Rare transmission on a fixed record)**.** *Assumptions. A finite record of patient contacts and worker identities, an initial source set and initially clean hands are fixed independently of transmission; there is no environment and no turnover. Pickup and transfer probabilities at contact i are pc_i and pd_i with fixed weights in $[0,1]$; contact trials are mutually independent and independent of clearing; the weights, the source law and the record are the same in both hygiene arms. Claim. The expected number of newly colonized patients satisfies*

$$\mathcal{I}_h(p) = p^2 K_h + O(p^3), \qquad K_h = \sum_{(i,j)\in\mathcal{B}} c_i d_j s_h(i,j), \tag{T4}$$

where $\mathcal{B}$ contains ordered pairs of a contact with an initial source and a later contact with an initially susceptible patient by the same worker, and $s_h(i,j)$ is the probability of no hand clearing or reset between them. If $K_{h_0} > 0$, then $1 - \mathcal{I}_{h_1}(p)/\mathcal{I}_{h_0}(p) \to 1 - K_{h_1}/K_{h_0}$ as $p \to 0$.

Proof. Preallocate independent pickup and transfer flags at every contact, used or not, with success probabilities pc_i and pd_i. For a fixed clearing record the expected number of acquisitions is a polynomial in p. With zero or one successful flag there is no acquisition, because all hands start clean. With exactly two successful flags, an acquisition occurs only when the first is an initial-source pickup and the second a later transfer to an initially susceptible patient by the same worker, with no clearing or reset between them; conversely each such surviving pair produces exactly one acquisition when all other flags fail. Its probability is $p^2 c_i d_j s_h(i,j)$ times a finite product of other flag-failure probabilities, which equals $1+O(p)$. Events with at least three successful flags have total probability $O(p^3)$ and bounded reward. Summing the eligible pairs proves (T4); an independent random clearing record or source law can be averaged without changing the argument, and dividing two expansions with $K_{h_0} > 0$ gives the limit. □

In the two-success configurations there was no earlier successful pickup, so the hand is clean at the source contact whatever the earlier completion states. If completion is stationary and independent of the contact record, the stationary bridge survival can therefore be used in K_h, and then $K_h = \sum_k A_k q_k$ with A_k the weighted number of bridges with k opportunities. This does not justify the stationary law at contamination times selected when p is appreciable, nor does the proof cover endogenous contact records or an initially dirty environment.

Small examples. For one worker with contact sequence ABAB, a uniform initial source among A and B and two stationary opportunities between consecutive contacts, $K_h = \frac{1}{2}\{3q_2(h,\rho,1) + q_6(h,\rho,1)\}$: three short and one long contributing pair, so two opportunities between adjacent contacts do not imply that every leading bridge has two. For two workers with sequence $A_u B_u B_v C_v$ and no clearing, the full model exceeds the source-only model by exactly $p^4/3$ under uniform sources A, B, C; the additional route needs four successes. A released exact checker evaluates four tiny fixed records at $p = 1/10, 1/100, 1/1000$ and ABAB under two completion laws (averaging all 64 six-opportunity completion paths) with exact fractions. These examples support the implementation and illustrate the proof; they do not establish long-horizon ICU accuracy.

T3.2 The fixed-horizon ICU expansion

Proposition T2 (Fixed-horizon ICU expansion)**.** *Assumptions. Hands and sites start clean; the tick horizon is finite (here 90+56 days); workers, beds and sites are finite and each worker makes*

finitely many contacts per tick, so there is a bounded number L of possible transmission slots; every pickup, transfer and exchange trial has probability pw with a state-dependent weight $w \in [0,1]$; other random draws (admissions, stays, schedules, hygiene) do not depend on p; hand contamination alone alters neither movement, screening, exposure nor the contact-location weights before an acquisition; and the outcome counts acquisitions by transfer during the observation window. Claim. With $\overline{K}_h = \mathbb{E}[K_h]$ the expected bridge coefficient over transmission-free backgrounds and $D^{(0)}$ their expected observation patient-days,

$$\mathbb{E}[C_h] = p^2 \overline{K}_h + O(p^3), \qquad \mathbb{E}[D_h] = D^{(0)} + O(p^2), \tag{T5}$$

so the leading-order rate is $R_h \simeq 1000 p^2 \overline{K}_h / D^{(0)}$ and, if $\overline{K}_{h_0} > 0$, $F \to F^* = 1 - \overline{K}_{h_1}/\overline{K}_{h_0}$ as $p \to 0$.

Proof. Pad unused slots and use at each slot an independent Bernoulli(p) gate followed by independent thinning with its state-dependent weight w; this reproduces the probability pw and the conditional transition law. Other draws use independent tapes. (This is a distributional construction, not a claim that changing p preserves a backend seed's event sequence.) With fewer than two open gates there is no acquisition. With exactly two, an acquisition requires an imported-carrier pickup and a later transfer to a susceptible patient by the same worker without clearing or reset. Before that transfer the patient and staff trajectory is the $p = 0$ background, because hand contamination alone changes neither movement, screening, exposure nor contact-location weights; later detection and relocation cannot create another acquisition with no remaining open gate. Only a transfer during observation counts, while its pickup may precede observation. Conditional on a transmission-free background, summing these pairs gives exactly $p^2(1-p)^{L-2}K_h$, and averaging over backgrounds replaces K_h by $\overline{K}_h$. Paths with three or more open gates have bounded reward and total probability $O(p^3)$. Patient-day exposure changes only after an acquisition, an $O(p^2)$ event, so the ratio of expected acquisition and exposure totals has leading term $p^2\overline{K}_h/D^{(0)}$; dividing the two arms gives the limit. □

For each pair the survival weight uses the actual preceding completion-state law and the ordered product of Section T2, unless a role reset kills the bridge; the first-ever opportunity uses the initialized Bernoulli(h) law and policy changes keep completion memory. In the two-gate paths there was no earlier pickup, so clean-hand selection does not alter this background law.

Remainder and higher-order routes. At a fixed horizon $\mathbb{E}[C_h]$ is a polynomial in p of degree at most L. Write $f(S)$ for the expected number of acquisitions when exactly the gates in S are open (an expectation over all other draws). Then

$$\mathbb{E}[C_h] = \sum_S p^{|S|}(1-p)^{L-|S|} f(S) = \sum_T p^{|T|} g(T), \qquad g(T) = \sum_{S \subseteq T} (-1)^{|T|-|S|} f(S),$$

by expanding $(1-p)^{L-|S|}$ (Möbius inversion). If T splits into nonempty parts A and B with $f(S) = f(S \cap A) + f(S \cap B)$ for every $S \subseteq T$, then $g(T) = 0$, because the alternating sum over the subsets of the other part vanishes; in particular $g(T) = 0$ whenever T contains a gate whose opening never changes the outcome (a pickup by an already contaminated hand, a contact with a clean object, a transfer to a colonized patient). The coefficient of p^m is therefore a sum over sets of m gates that interact: a second pickup on the same hand, two bridges competing for one susceptible patient, hand–site–hand chains, or an acquired patient acting as a source. The factor $(1-p)^{L-2}$ in the exactly-two term, which is numerically negligible for L of order 10^6, does not by itself give a useful error scale: the remainder still depends on L and the horizon through how many interacting

gate sets accompany each bridge, and $p^2\overline{K}_h$ captures 25–60% of the measured control rates in the main text. We did not compute the p^3 or p^4 coefficients, so no useful finite-p error bound has been established for 146 days of ICU traffic, and the limit is used as a structural diagnostic. A clean-site environmental route needs pickup, deposition, repickup and transfer, four successes, and a newly acquired patient used as a new source also needs four. Feedback is nevertheless not generally fourth order: the original dirty hand can colonize B on its second success and C on its third; allowing a C contact only after B acquires gives exactly $p^2 + p^3$ expected acquisitions without clearing. Dirty initial sites would add two-success environmental bridges, and dirty initial hands an $O(p)$ term. The constant in the remainder depends on the horizon; no uniform long-run accuracy, equilibrium or omission bound follows.

T3.3 Calibration scale and response ratio

Fitting a baseline rate determines $p^2\overline{K}_{h_0}$ at leading order and leaves the ratio $\overline{K}_{h_1}/\overline{K}_{h_0}$ free. Two constructions make this precise.

Exact equal-baseline nonidentification. One initially clean worker contacts a known carrier, then crosses exactly k opportunities, then contacts a susceptible patient; pickup and transfer succeed independently with probability p and there is no other route. Write $s_j(\theta) = q_k(h_j;\theta)$ for the survival through the k opportunities in arm j under mechanism θ. The acquisition probability is exactly $I_j(p) = p^2 s_j(\theta)$. For mechanisms with $s_0(\theta) > 0$ and any target $0 < b \le \min_\theta s_0(\theta)$, the choice $p_\theta = \sqrt{b/s_0(\theta)} \in (0,1]$ gives every mechanism baseline acquisition probability exactly b and the same mean completion h_0, yet $F_\theta = 1 - s_1(\theta)/s_0(\theta)$. For $k = 2$ and .90→.95 the three profiles give exactly 3/4, 23/44 and 13/49 (75.0%, 52.3% and 26.5%), and any $b \le .01$ permits all three calibrations. The baseline observation map is not injective on this model class, even without noise. Each arm starts from the stationary law, not the ICU's transient switch.

Lemma T1 (Local calibration branch)**.** *Assumptions. Expected baseline and policy acquisitions satisfy $I_0(p) = K_0p^2 + O(p^3)$ and $I_1(p) = K_1p^2 + O(p^3)$ with $K_0 > 0$, and are analytic near $p = 0$. Under the fixed-horizon gate construction of Proposition T2 they are polynomials in p of degree at most L, so this holds automatically; for other models an $O(p^3)$ remainder alone would not give local monotonicity. Claim. For sufficiently small targets $b > 0$ there is a unique local calibration branch $p(b) = \sqrt{b/K_0} + O(b)$, and $F(p(b)) = 1 - K_1/K_0 + O(\sqrt{b})$.*

Proof. $I_0'(p) = 2K_0p + O(p^2) > 0$ for small $p > 0$, so I_0 is invertible near zero; writing $I_0 = p^2(K_0 + O(p))$ gives $p = O(\sqrt{b})$ and the expansion of the positive square root. The policy-to-baseline ratio is $K_1/K_0 + O(p)$. □

Two mechanisms whose limiting ratios differ therefore have different exact-equal-baseline responses for all sufficiently small targets. The lemma gives no numerical threshold showing that 3 per 1,000 patient-days is small enough, and no finite-p error bound.

T3.4 Identification of a policy response from bridge features

Let J be a finite set of bridge types (a type may record opportunity count, timing, worker, reset history and initial completion-state law). *Assume* that each type's nonnegative contact/source coefficient A_j is unchanged by the policy, that baseline and policy survival s_{0j}, s_{1j} are known, and that $s_{0j} > 0$ and $K_0 = \sum_j A_j s_{0j} > 0$. (A type with $s_{0j} = 0 < s_{1j}$ is invisible in the normalization;

if policy survival never exceeds baseline survival such types contribute to neither coefficient and can be removed.) Define the survival-weighted shares and survival ratios

$$w_j = \frac{A_j s_{0j}}{K_0}, \qquad r_j = \frac{s_{1j}}{s_{0j}}, \qquad \kappa := \frac{K_1}{K_0} = \sum_j w_j r_j, \qquad F = 1 - \kappa. \tag{T6}$$

Suppose additional observations specify linear features $Mw = m$, the first row of M being all ones, with feasible set $\mathcal{W}(m) = \{w \geq 0 : Mw = m\} \neq \varnothing$ and feasible support $J_m = \{j : \text{some } w \in \mathcal{W}(m) \text{ has } w_j > 0\}$.

Proposition T3 (Identification criterion)**.** *κ takes the same value for every $w \in \mathcal{W}(m)$ if and only if $r_{J_m} \in \mathrm{row}(M_{J_m})$, where M_J restricts M to the columns in J. If some feasible w is positive in every coordinate, the condition is $r \in \mathrm{row}(M)$.*

Proof. If $r_{J_m} = M_{J_m}^\top \lambda$, then $r^\top w = \lambda^\top m$ for every feasible w, since coordinates outside J_m vanish. Conversely, if the condition fails, there is v supported on J_m with $Mv = 0$ and $r^\top v \neq 0$. Averaging feasible vectors that are positive in each coordinate of J_m gives a feasible w^* positive on all of J_m; for small $\epsilon > 0$ both $w^* \pm \epsilon v$ are feasible, and their ratios differ by $2\epsilon r^\top v$. □

Baseline normalization alone ($M =$ the row of ones) identifies the response only when the survival ratio is constant over the feasible bridge types. The feasible-support qualification matters: at a boundary, constraints can force a single distribution even when r is outside the unrestricted row space. The criterion can also be applied directly to the nonnegative coefficients A, with observed raw moments and the baseline coefficient as constraint rows; raw (contact-weighted) and survival-weighted moments are different observations.

T3.5 Sharp bounds

Free spectrum. If nothing beyond normalization is known and $q_{0k} > 0$ on a finite support, then $1 - \max_k r_k \leq F \leq 1 - \min_k r_k$, sharp over nonnegative coefficient spectra with free scale ($A_k = cv_k/q_{0k}$ realizes any share vector v). Sharpness over this algebraic class does not mean that an extremal spectrum is realizable by a specified ICU schedule. For $\rho \geq 0$ the completion kernel is monotone (completion is at least as likely after a completion as after a miss), so under a common-uniform coupling of the stationary chains, completions under $h_1 \geq h_0$ dominate those under h_0, so $q_{1k} \leq q_{0k}$ and $0 \leq F \leq 1$; zero-opportunity bridges have $r_0 = 1$.

Linear information. If additional information constrains the shares by $Mv = d$ and $Nv \leq u$, the sharp lower and upper ratios solve the linear programs $\min / \max\, r^\top v$ over the constrained simplex. Valid added measurements can only shrink the feasible set. With m independent moment equalities plus normalization and no active inequalities, an optimum has at most $m + 1$ positive weights (extreme points of the feasible polytope). These are value-of-information calculations, not confidence intervals; they become confidence statements only if the constraints come from a joint confidence set, whose projection preserves its coverage under the stated model. If information is on raw coefficients A instead, K_1/K_0 is a linear-fractional program; with $x = A/K_0$ and $t = 1/K_0$ it becomes linear ($q_0^\top x = 1$, $Mx = dt$, $Nx \leq ut$, objective $q_1^\top x$).

Proposition T4 (One-moment bounds under independent completion)**.** *Assumptions. Independent completion with fixed $0 < e \leq 1$, one opportunity rule, a finite spectrum supported on all integers $k_{\min}, \ldots, k_{\max}$ unchanged by h, and $0 < b < a$ with $a = 1 - h_0 e$, $b = 1 - h_1 e$, $s = b/a$.*

Then $\kappa = \mathbb{E}_w[s^K]$, a probability generating function of the survival-weighted opportunity count. Let $\mu = \mathbb{E}_w[K]$, $\ell = \lfloor \mu \rfloor$, $\theta = \mu - \ell$, $B_-(\mu) = (1-\theta)s^\ell + \theta s^{\ell+1}$ (with $B_- = s^{k_{\max}}$ at $\mu = k_{\max}$) and $B_+(\mu) = \frac{k_{\max}-\mu}{k_{\max}-k_{\min}} s^{k_{\min}} + \frac{\mu - k_{\min}}{k_{\max}-k_{\min}} s^{k_{\max}}$. Claim. $1 - B_+(\mu) \le F \le 1 - B_-(\mu)$, and both bounds are attained.

Proof. $x \mapsto s^x$ is convex, so its chord between $k_{\min}$ and $k_{\max}$ lies above it and $\mathbb{E}[s^K] \le B_+(\mu)$, with equality for mass $(k_{\max} - \mu)/(k_{\max} - k_{\min})$ at $k_{\min}$ and the rest at $k_{\max}$ (the Edmundson–Madansky bound). The piecewise-linear interpolation of s^k on the integers is convex because its slopes increase, so Jensen's inequality gives $\mathbb{E}[s^K] \ge B_-(\mu)$, with equality for mass $1 - \theta$ at ℓ and θ at $\ell + 1$. □

Example. With h .90→.95, $e = 1$, support $\{2, \dots, 6\}$ and $\mu = 4$, $s = 1/2$ and $111/128 \le F \le 15/16$ (86.7% to 93.8%); both endpoints have the same weighted mean. Expanding (T6) with $z = e(h_1 - h_0)/a$ gives $F = 1 - \mathbb{E}_w[(1-z)^K]$ and $F^{(r)}(0) = (-1)^{r+1}(e/a)^r \mathbb{E}_w[(K)_r]$, so the weighted mean identifies the local slope $F'(0) = e\mu/(1 - h_0 e)$ but generally not a finite improvement; $0 \le \mu z - F \le \frac{1}{2}\mathbb{E}_w[K(K-1)]z^2$, which can be loose ($z = 1/2$ here). For stationary Markov completion with perfect clearing and $k \ge 1$, the ratio is $c\,d^{k-1}$ with $c = (1 - h_1)/(1 - h_0)$ and $d = [1 - (1-\rho)h_1]/[1 - (1-\rho)h_0]$, and the same convexity argument applies for known ρ.

Raw versus survival-weighted information. Raw bridge proportions are not the shares v_k. With independent perfect clearing, .90→.95 and support $\{2, 6\}$, $r_2 = .25$ and $r_6 = .015625$, so without further information $.75 \le F \le .984375$; knowing $v_6 \in [.1, .3]$ gives $F = .75 + .234375 v_6 \in [.7734, .8203]$. By contrast, raw six-opportunity proportions from .5 to .9 give only $F \in [.750023, .750211]$, because $q_{02} = .01$ while $q_{06} = 10^{-6}$: survival weighting, not bridge counting, is the relevant transformation.

Same baseline and same raw mean, different response. With independent completion, perfect clearing and $k = 1, \dots, 4$, the spectra $A^- = (11/42, 3/28, 1/2, 11/84)$ and $A^+ = (5/21, 11/28, 0, 31/84)$ both have total 1, raw mean 5/2 and $K_{.90} = 1111/40000$, yet $F^- = 192845/373296 \approx .5166$ and $F^+ = 200065/373296 \approx .5359$ for .90→.95. In direct-A form the constraint rows 1, k and $(1/10)^k$ have rank three and adjoining the policy row $(1/20)^k$ raises it to four, so by Proposition T3 the response is not identified. Four disjoint two-contact modules (pickup probability pA_k, transfer probability p) realize these spectra exactly for every $p \in [0, 1]$, so the two models have exactly equal baseline expected acquisitions at the same p. This establishes logical nonidentification, not attainability in the ICU.

T3.6 Identification from (h, q_1, q_2)

Proposition T5 (Identification of the stationary hygiene law)**.** *Assumptions. Stationary two-state Markov completion with known mean $0 < h < 1$ and lag-one correlation ρ; independent clearing trials with constant probability e given completion; contamination initialized independently of the completion state (as in a controlled assay). Claim. If $e > 0$,*

$$e = \frac{1 - q_1}{h}, \qquad \rho = \frac{q_2 - q_1^2}{e^2 h(1-h)}, \tag{T7}$$

so (h, q_1, q_2) identify (ρ, e) and hence every stationary q_k. A policy response additionally requires a fixed, known contact structure and unchanged ρ, e under policy.

Proof. Invert (T3): $q_1 = 1 - he$ gives e, and $q_2 - q_1^2 = e^2 \rho h(1-h)$ gives ρ. □

At $e = 0$, ρ is not identified by survival, but then $q_k = 1$ and the hand-only effect is zero. At $h = .90$ the primary profiles have $(q_1, q_2) = (.10, .0100)$, $(.10, .0550)$ and $(.28, .0784)$: q_1 separates perfect from imperfect clearing but cannot separate independent from clustered misses, which q_2 does. Valid estimates must obey the family's feasibility constraints rather than being clipped. The result is conditional: q_1 and q_2 do not identify q_3 or longer survival of an arbitrary (non-Markov) process (Section T11 gives a prespecified counterexample with matching q_1, q_2), nor nonstationary initial states or a change in ρ or e caused by the intervention. Survival among naturally occurring contamination episodes need not estimate the stationary q_k, because episode selection can depend on preceding completion states; possible designs are an assay that initializes contamination independently of completion, or estimation of ρ from unselected completion records and of e from paired pre/post contamination measurements under a constant-e model.

T3.7 Serial dependence reverses its effect with clearing efficacy

Proposition T6 (Sign change for a two-opportunity bridge)**.** *Assumptions. A stationary bridge with exactly two opportunities,* $0 < h_0 < h_1 < 1$, $0 < e \le 1$ *and* $\rho \in [0, 1]$*, with* ρ *and* e *unchanged by the policy. Write* $a_j = (1 - h_j e)^2$, $b_j = e^2 h_j (1 - h_j)$ *and* $\kappa_2(\rho, e) = (a_1 + \rho b_1)/(a_0 + \rho b_0)$, $F = 1 - \kappa_2$*. Claim.*

$$\frac{\partial F}{\partial \rho} = \frac{b_0 a_1 - b_1 a_0}{(a_0 + \rho b_0)^2}, \tag{T8}$$

whose sign does not depend on ρ*. For* $h_0 = .90$, $h_1 = .95$ *the unique threshold in* $(0, 1)$ *is*

$$e^* = \frac{\sqrt{h_0(1-h_0)} - \sqrt{h_1(1-h_1)}}{h_1\sqrt{h_0(1-h_0)} - h_0\sqrt{h_1(1-h_1)}} = 0.92352808870\ldots; \tag{T9}$$

for $e < e^*$ *clustering raises relative prevention and for* $e > e^*$ *it lowers it. More generally, for* $0 < h_0 < h_1 < 1$ *a threshold* $e^* \in (0, 1)$ *exists, and is then unique, if and only if* $h_0 + h_1 > 1$*; otherwise clustering lowers relative prevention at every* $e \in (0, 1]$*.*

Proof. Differentiate the ratio. The numerator $b_0 a_1 - b_1 a_0$ vanishes when $e\sqrt{h_0(1-h_0)}(1 - h_1 e) = e\sqrt{h_1(1-h_1)}(1 - h_0 e)$ (both factors positive), which is linear in e after division by e and gives e^*; the sign on each side follows by evaluation at $e = 1$ (negative) and $e \to 0^+$ (positive) for these h values. For the general statement, $b_1/b_0 = h_1(1-h_1)/\{h_0(1-h_0)\}$ does not depend on e, whereas $a_1/a_0 = \{(1 - h_1 e)/(1 - h_0 e)\}^2$ decreases strictly in e (its logarithmic derivative is $2(h_0 - h_1)/\{(1 - h_0 e)(1 - h_1 e)\} < 0$), from 1 at $e = 0$ to $\{(1 - h_1)/(1 - h_0)\}^2$ at $e = 1$. The latter is always below b_1/b_0: dividing both by $(1 - h_1)/(1 - h_0)$ leaves $(1 - h_1)/(1 - h_0) < h_1/h_0$, whose left side is below 1 and right side above 1 when $h_1 > h_0$. Hence $a_1/a_0 - b_1/b_0$ is negative at $e = 1$ and crosses zero exactly once in $(0, 1)$ if and only if its value near $e = 0$, $1 - b_1/b_0$, is positive, that is, if and only if $h_1(1-h_1) < h_0(1-h_0)$, which for $h_0 < h_1$ is $h_0 + h_1 > 1$. Otherwise $a_1/a_0 < b_1/b_0$ on all of $(0, 1]$ and $\partial F_2/\partial \rho < 0$. □

At $\rho = 0, .5, 1$ the exact prevention is 3/4, 23/44, 1/2 at $e = 1$ and 13/49, 43/134, 6/17 at $e = .8$ (75.0%, 52.3%, 50.0%; 26.5%, 32.1%, 35.3%). The interior comparison of $\rho = 0$ and .5 already shows the reversal without the reducible $\rho = 1$ case.

Completion range. Table T6 gives e^* and the isolated two-opportunity responses for other completion ranges. The threshold falls as the range moves down ($e^* = .47$ for 60%→70%, .83 for 80%→90%) and does not exist for $h_0 + h_1 \le 1$, so on an isolated two-opportunity bridge clustering raises relative prevention only when $h_0 + h_1 > 1$ and $e < e^*$. On the isolated bridge, clustered

misses and imperfect clearing nearly tie at 50%→60% and swap order at 40%→50%. The full ICU coefficient does not follow the isolated thresholds: at $e = .8$ its clustering contrast is already positive at 70%→80% (+0.254 points, where the isolated bridge gives −2.383) and is negative at 60%→70% (Section T4.4).

Table T6: Clearing threshold e^* and isolated two-opportunity prevention F_2 (%) for several completion ranges (closed forms; not full-ICU limits). Combined: $\rho = .5$, $e = .8$.

$h_0 \to h_1$	$h_0 + h_1$	e^*	Independent	Clustered	Imperfect	Combined
0.40→0.50	0.90	none	30.6	21.9	22.1	18.4
0.50→0.60	1.10	0.184	36.0	25.3	24.9	21.1
0.60→0.70	1.30	0.465	43.8	30.4	28.4	24.9
0.70→0.80	1.50	0.673	55.6	38.5	33.1	30.7
0.80→0.90	1.70	0.833	75.0	54.2	39.5	40.7
0.90→0.95	1.85	0.924	75.0	52.3	26.5	32.1
0.95→0.99	1.94	0.977	96.0	80.8	24.9	36.2

Longer bridges. The ρ-independence of the sign holds because q_2 is affine in ρ; q_k is a polynomial of degree $k-1$ in ρ, and isolated F_k is not monotone in ρ at some e for $k = 3$, 4 and 6 (for example $e \in [.83, .92]$ at $k = 3$). For isolated stationary bridges of k opportunities at .90→.95, $\Delta_k(e) = F_k(\rho{=}.5, e) - F_k(0, e)$ changes sign at $e = .9235$, .881, .845 and .789 for $k = 2$, 3, 4 and 6. These values show that bridge length shifts the crossing, and a crossing exists at every tested length. The ICU coefficient's opportunity-count mix is reported in Table T13.

Proposition T7 (Local sensitivity of a stationary bridge spectrum)**.** *Assume stationary Markov completion without resets, $0 < h_0 < h_1 < 1$ and $0 < e \le 1$, with $\rho \in [0, 1)$. Let $K_i(\rho) = \sum_{k \ge 0} A_k q_k(h_i, \rho, e)$, where the nonnegative weights A_k have finite support, are common to both arms and all ρ, and are not all zero. Define the comparator response $F_{\text{spec}} = 1 - K_1/K_0$. Put*

$$u_i = 1 - h_i e, \qquad \beta_i = e^2 h_i (1 - h_i), \qquad \tau_i = \beta_i / u_i^2, \qquad \ell(k) = \max\{k - 1, 0\},$$

$$w_k = \frac{A_k u_0^k}{K_0(0)}, \qquad \kappa = \frac{K_1(0)}{K_0(0)}, \qquad v_k = \frac{w_k (u_1/u_0)^k}{\kappa}.$$

Then the right derivative at independence is

$$\left.\frac{\partial F_{\text{spec}}}{\partial \rho}\right|_{0^+} = \kappa\{\tau_0 \mathbb{E}_w \ell - \tau_1 \mathbb{E}_v \ell\}. \tag{T10}$$

The bracket therefore determines the local sign. The condition $\tau_0 \ge \tau_1$ is sufficient for a nonnegative derivative; it is strictly positive if $\tau_0 > \tau_1$ and some mass lies at $k \ge 2$. When $h_0 + h_1 > 1$, this sufficient condition is $e \le e_2^*$, where e_2^* is the threshold e^* in Proposition T6. It is not necessary for a mixture.

Proof. Write $P_\rho = \mathbf{1}\pi + \rho(I - \mathbf{1}\pi)$. Differentiating the $k-1$ transitions in the stationary expression $q_k = \pi\Gamma(P_\rho\Gamma)^{k-1}\mathbf{1}$ at zero leaves one adjacent centered pair, contributing β_i, and $k-2$ independent factors, contributing u_i^{k-2}. Thus $q_k'(0) = (k-1)\beta_i u_i^{k-2}$ for $k \ge 2$, and zero for $k = 0, 1$. Finite support permits differentiation term by term. Differentiating the ratio gives Eq. (T10). The distribution v is the decreasing exponential tilt of w by $(u_1/u_0)^k$, so $\mathbb{E}_v \ell \le \mathbb{E}_w \ell$: explicitly, the covariance of increasing $\ell(k)$ with the decreasing tilt is nonpositive. The sufficient condition and its strict version follow. For a spectrum supported on $k = 0, 1$, the derivative is zero. Writing

$s_i = \sqrt{h_i(1-h_i)}$, the condition is $s_0 - s_1 \geq e(h_1 s_0 - h_0 s_1)$; the coefficient of e is positive. At high completion this gives exactly the e_2^* of Proposition T6. A mixture can satisfy the bracket condition even if $\tau_0 < \tau_1$, because its two tilted means differ. □

Proposition T8 (Crossing for every fixed bridge length)**.** *Let $0 < h_0 < h_1 < 1$ with $h_0 + h_1 > 1$. Under stationary Markov completion without resets, for every fixed integer $k \geq 2$ and every fixed $\rho \in (0,1)$, the difference $F_k(\rho, e) - F_k(0, e)$ has at least one sign-changing root in $e \in (0,1)$. It is positive for sufficiently small positive e and negative at $e = 1$. No uniqueness or common crossing across bridge lengths is asserted.*

Proof. For fixed k and ρ, put $S_k(\rho) = \sum_{d=1}^{k-1}(k-d)\rho^d > 0$. The stationary identity $\mathbb{E}[X_i X_j] = h^2 + h(1-h)\rho^{j-i}$ gives the finite-product expansion

$$q_k(h, \rho, e) = 1 - khe + \left\{ \binom{k}{2} h^2 + h(1-h)S_k(\rho) \right\} e^2 + O(e^3).$$

Expanding the survival quotient and subtracting its independent value cancels the ρ-independent terms, giving

$$F_k(\rho, e) - F_k(0, e) = (h_1 - h_0)(h_0 + h_1 - 1)S_k(\rho)e^2 + O(e^3) > 0$$

for sufficiently small positive e. This neighborhood can depend on k and ρ. At $e = 1$, write $m_i = 1 - h_i$ and $c_i = \rho + (1-\rho)m_i$. Then $q_k(h_i, \rho, 1) = m_i c_i^{k-1}$ and $c_1/c_0 > m_1/m_0$, so $F_k(\rho, 1) - F_k(0, 1) < 0$. Both baseline survival denominators are strictly positive on $[0, 1]$. The difference is a rational function of e with a nonzero polynomial numerator, hence has finitely many roots. Continuity and the opposite signs therefore imply at least one root where the sign changes. □

For each single $k \geq 2$, Proposition T7 also puts the zero-correlation derivative's sign change at e_2^*. A finite-ρ crossing need not coincide with that local threshold.

Counterexample for arbitrary spectra. Take $A_0 = A_2 = 1$, all other weights zero, and $h_0 = .9$, $h_1 = .95$. With $u_0 = 1 - .9e$ and $\beta_0 = .09e^2$, direct subtraction gives

$$F_{\text{spec}}(\rho, e) - F_{\text{spec}}(0, e) = \frac{\rho e^2 \{171(e-1)^2 + 169\}}{4000(1 + u_0^2)(1 + u_0^2 + \rho\beta_0)} > 0$$

for every $e \in (0, 1]$, $\rho \in (0, 1)$. This finite, nonnegative spectrum has no efficacy crossing. It is a comparator counterexample, not an asserted ICU bridge distribution. The stationary, no-reset propositions do not establish a global crossing theorem for arbitrary mixtures or the reset ICU engine.

In the full ICU, at leading order, the same reversal appears on the computed surface: $\Delta(e) = F^*(\rho{=}.5, e) - F^*(0, e)$ is -21.5 percentage points at $e = 1$ and $+9.0$ at $e = .8$. On the original-workflow fine grid of the prespecified extension (Section T4.4), $\Delta(e)$ is strongly nonlinear, largest near $e = .8$, and changes sign once, between .93 and .94 (.93361 by linear interpolation; smooth fits give .9338, post hoc, Section T4.4). This lies above the isolated two-opportunity root .92353 and close to the renewal comparator's .936; the two arms weight bridge lengths differently, so per-length isolated roots do not bound it. Its narrow bootstrap interval describes background variation only and does not bound interpolation error. It is not a finite-p ICU result: bridge-length mixtures, transient completion states, surfaces and feedback can change the pattern.

Exact checks. An exact-rational checker passed 229 assertion groups covering the early formulas, inverse identification, counterexamples and sharp bounds; a separate generator recomputed the displayed fractions. Later spectrum and fixed-length results were checked by exact enumeration of binary histories. The proofs establish the general statements. These internal checks neither replace proofs nor validate the ICU comparator assumptions; exact arithmetic, case counts and verification details are in the repository execution records.

T4 Coefficient computation and the extended surface in both workflows

T4.1 Coefficient computation

A background ω is a $p = 0$ history under Table T2: 90 warmup and 56 observation days, 3% imported carriage and care-zone opportunities. Without transmission, hygiene cannot change movement, screening or isolation, so both arms use one history. The engine sums every bridge from an imported carrier contact i to the same worker's susceptible-patient contact j in the observation window, with no intervening role reset. Warmup pickups are included. Each bridge contributes $c_i d_j s_h(i,j)$, with location weights $c_i, d_j \in \{1, .6\}$ and exact survival $\nu_i \prod_m [P(h_m, \rho)\Gamma(e_m)]\mathbf{1}$ from the actual preceding completion-state law. The .90→.95 switch retains completion memory. Their sum is $K_h(\omega)$; Proposition T2 identifies $\mathbb{E}[K_h]$ as the p^2 acquisition coefficient.

The limit is $F^* = 1 - \overline{K}_{h_1}/\overline{K}_{h_0}$, a ratio of means with no background dropped. The 256 backgrounds are split into two banks of 128. Seed-level normal K intervals and covariance-preserving normal/delta F^* intervals are reported. Workflow differences use 20,000 whole-seed bootstrap draws with one index matrix (NumPy seed 20261021). Hygiene survival is integrated exactly; intervals cover background variation only. Strong within-history covariance makes them narrow. They describe neither staff-behaviour uncertainty nor finite-p approximation error.

Table T7: Primary leading-order coefficients on the same 256 transmission-free backgrounds in both workflows (completion .90→.95). K in model units with seed-level normal 95% intervals; F^* in percent with normal/delta and whole-seed bootstrap 95% intervals. These are Monte Carlo intervals over backgrounds only.

Profile	Workflow	Mean K_0 [95%]	Mean K_1 [95%]	F^* (%) [normal/delta 95%]	F^* bootstrap 95% interval
Independent misses	Original	55.51 [53.1, 58.0]	13.80 [13.2, 14.4]	75.13 [75.10, 75.17]	[75.09, 75.16]
	Alternative	53.37 [51.1, 55.6]	13.25 [12.7, 13.8]	75.18 [75.18, 75.18]	[75.18, 75.18]
Clustered misses	Original	409.13 [391.0, 427.2]	189.65 [181.3, 198.0]	53.65 [53.64, 53.65]	[53.64, 53.65]
	Alternative	393.16 [376.4, 409.9]	182.22 [174.5, 190.0]	53.65 [53.65, 53.65]	[53.65, 53.65]
Imperfect clearing	Original	464.91 [444.3, 485.5]	334.77 [319.9, 349.6]	27.99 [27.99, 28.00]	[27.99, 28.00]
	Alternative	447.05 [428.0, 466.1]	321.94 [308.2, 335.7]	27.99 [27.98, 27.99]	[27.98, 27.99]

T4.2 The extended surface

The surface evaluates the 16 (ρ, e) pairs in $\{0, .25, .5, .75\} \times \{1, .9, .8, .6\}$ at .90→.95 completion and the three primary profiles at .80→.90 and .95→.99, 22 pairs in all, on the same backgrounds. Each pair keeps its own baseline warmup law and completion memory across the switch. Tables T8 and T9 report both workflows.

Table T8: Leading-order coefficients for all 22 pairs in both workflows (256 shared backgrounds). Original and alternative workflows are defined in Table T3.

			Original workflow			Alternative workflow		
$h_0 \to h_1$	ρ	e	K_0	K_1	F^* (%)	K_0	K_1	F^* (%)
0.90→0.95	0.00	1.0	55.51	13.80	75.13	53.37	13.25	75.18
0.90→0.95	0.00	0.9	205.48	117.94	42.60	197.63	113.43	42.61
0.90→0.95	0.00	0.8	464.91	334.77	27.99	447.05	321.94	27.99
0.90→0.95	0.00	0.6	1429.79	1216.37	14.93	1374.18	1169.13	14.92
0.90→0.95	0.25	1.0	197.89	85.63	56.73	190.24	82.31	56.73
0.90→0.95	0.25	0.9	333.15	181.54	45.51	320.28	174.54	45.50
0.90→0.95	0.25	0.8	580.57	391.95	32.49	558.11	376.86	32.48
0.90→0.95	0.25	0.6	1525.28	1263.08	17.19	1465.90	1213.99	17.18
0.90→0.95	0.50	1.0	409.13	189.65	53.65	393.16	182.22	53.65
0.90→0.95	0.50	0.9	530.53	278.25	47.55	509.86	267.42	47.55
0.90→0.95	0.50	0.8	765.71	482.31	37.01	735.96	463.64	37.00
0.90→0.95	0.50	0.6	1689.06	1342.21	20.54	1623.40	1290.06	20.53
0.90→0.95	0.75	1.0	893.86	423.37	52.64	859.67	407.12	52.64
0.90→0.95	0.75	0.9	1001.18	504.56	49.60	962.91	485.23	49.61
0.90→0.95	0.75	0.8	1223.71	701.95	42.64	1176.96	675.08	42.64
0.90→0.95	0.75	0.6	2125.66	1550.51	27.06	2044.11	1490.76	27.07
0.80→0.90	0.00	1.0	228.54	55.51	75.71	219.80	53.37	75.72
0.80→0.90	0.00	0.8	807.56	464.91	42.43	776.34	447.05	42.42
0.80→0.90	0.50	1.0	953.96	409.15	57.11	916.79	393.17	57.11
0.95→0.99	0.00	1.0	13.80	0.59	95.74	13.25	0.53	96.01
0.95→0.99	0.00	0.8	334.77	247.98	25.93	321.94	238.49	25.92
0.95→0.99	0.50	1.0	189.64	35.73	81.16	182.21	34.30	81.18

Table T9: Paired workflow differences on the same backgrounds, the isolated two-opportunity comparator and the renewal approximation. $\Delta F^* = F^*_{\text{alt}} - F^*_{\text{orig}}$ in percentage points with paired whole-seed bootstrap 95% intervals; K_0 change is the relative change of the mean baseline coefficient with its bootstrap 95% interval. The q_2 column is $1 - q_2(h_1)/q_2(h_0)$ for a stationary bridge with exactly two opportunities; it has no Monte Carlo interval and is shown for comparison only. F_{geo} is the renewal approximation of Section T4.2 (post hoc).

$h_0 \to h_1$	ρ	e	F^*_{orig}	F^*_{alt}	ΔF^* (pp) [95%]	K_0 change (%) [95%]	q_2 (%)	F_{geo} (%)
0.90→0.95	0.00	1.0	75.13	75.18	+0.051 [+0.019, +0.090]	−3.9 [−6.8, −0.8]	75.00	75.19
0.90→0.95	0.00	0.9	42.60	42.61	+0.001 [−0.005, +0.007]	−3.8 [−6.8, −0.8]	41.76	42.66
0.90→0.95	0.00	0.8	27.99	27.99	−0.008 [−0.012, −0.004]	−3.8 [−6.8, −0.8]	26.53	28.15
0.90→0.95	0.00	0.6	14.93	14.92	−0.005 [−0.010, −0.001]	−3.9 [−6.8, −0.9]	12.62	15.48
0.90→0.95	0.25	1.0	56.73	56.73	+0.005 [−0.002, +0.014]	−3.9 [−6.8, −0.8]	55.77	56.88
0.90→0.95	0.25	0.9	45.51	45.50	−0.006 [−0.012, 0.000]	−3.9 [−6.8, −0.8]	43.59	45.77
0.90→0.95	0.25	0.8	32.49	32.48	−0.012 [−0.019, −0.005]	−3.9 [−6.8, −0.8]	29.74	32.93
0.90→0.95	0.25	0.6	17.19	17.18	−0.006 [−0.013, +0.001]	−3.9 [−6.8, −0.9]	13.89	18.03
0.90→0.95	0.50	1.0	53.65	53.65	+0.004 [0.000, +0.009]	−3.9 [−6.9, −0.9]	52.27	54.04
0.90→0.95	0.50	0.9	47.55	47.55	−0.003 [−0.009, +0.004]	−3.9 [−6.8, −0.9]	44.50	48.24
0.90→0.95	0.50	0.8	37.01	37.00	−0.009 [−0.019, +0.001]	−3.9 [−6.8, −0.9]	32.09	38.12
0.90→0.95	0.50	0.6	20.54	20.53	−0.002 [−0.012, +0.008]	−3.9 [−6.8, −0.9]	15.08	22.04
0.90→0.95	0.75	1.0	52.64	52.64	+0.006 [+0.003, +0.009]	−3.8 [−6.8, −0.8]	50.81	53.07
0.90→0.95	0.75	0.9	49.60	49.61	+0.005 [−0.001, +0.010]	−3.8 [−6.8, −0.8]	45.05	50.47
0.90→0.95	0.75	0.8	42.64	42.64	+0.004 [−0.008, +0.015]	−3.8 [−6.8, −0.8]	33.88	44.39
0.90→0.95	0.75	0.6	27.06	27.07	+0.013 [−0.003, +0.029]	−3.8 [−6.8, −0.8]	16.18	29.84
0.80→0.90	0.00	1.0	75.71	75.72	+0.007 [−0.001, +0.017]	−3.8 [−6.8, −0.8]	75.00	75.76
0.80→0.90	0.00	0.8	42.43	42.42	−0.014 [−0.021, −0.007]	−3.9 [−6.8, −0.8]	39.51	42.87

$h_0 \to h_1$	ρ	e	F^*_{orig}	F^*_{alt}	ΔF^* (pp) [95%]	K_0 change (%) [95%]	q_2 (%)	F_{geo} (%)
0.80→0.90	0.50	1.0	57.11	57.11	+0.003 [−0.002, +0.009]	−3.9 [−6.8, −0.9]	54.17	57.95
0.95→0.99	0.00	1.0	95.74	96.01	+0.272 [+0.108, +0.471]	−4.0 [−7.0, −1.0]	96.00	96.01
0.95→0.99	0.00	0.8	25.93	25.92	−0.005 [−0.008, −0.001]	−3.8 [−6.8, −0.8]	24.89	26.01
0.95→0.99	0.50	1.0	81.16	81.18	+0.016 [+0.006, +0.028]	−3.9 [−6.9, −0.9]	80.76	81.29

Coefficient-surface results. Increasing ρ lowers prevention at $e = 1$ and raises it at $e = .9, .8, .6$ (Table T8). For $\Delta(e) = F^*(.5, e) - F^*(0, e)$, the fine-grid crossing is about .934 (Section T4.4). The earlier .9-to-1 interpolation understates this nonlinear crossing; historical prediction scores retain that protocol statistic.

The isolated two-opportunity comparator works best for independent perfect clearing, where about 99% of baseline K has two opportunities. Longer bridges matter elsewhere: the largest displayed comparator gap is 10.9 points. Opportunity-count mixtures were recorded only for the three primary profiles (Table T13). Other surface rows have comparator values but no measured mixtures.

A post hoc renewal approximation assumes stationary, unbounded contamination with a susceptible contact every second opportunity. Then $S(h) = \sum_{j \geq 1} q_{2j}(h) = \pi \Gamma P \Gamma \{I - (P\Gamma)^2\}^{-1} \mathbf{1}$ and $F_{\text{geo}} = 1 - S(h_1)/S(h_0)$. It exceeds F^* in all 22 rows, with maximum error 2.8 points (13 rows within .5 and 18 within 1), and crosses at $e = .936$ for $\rho = .5$. The expanded grid fails its prespecified error bound (Section T4.4). This discrete analogue of a worker contamination period (Austin et al., 1999) depends only on $q_1 = 1 - he$ under memoryless clearing, but not on (h, q_1) alone under Markov completion. Its fit may reflect two-opportunity spacing, resets and the nearly uniform small-even-count raw spectrum (Table T11); it is not a theorem.

The primary ordering independent > clustered > imperfect holds in all computed ICU completion ranges (Tables T10 and T17), descriptively including 60–70% and 70–80%. It fails for the isolated bridge at 40–50%, where imperfect clearing narrowly exceeds clustering (Table T6); no ICU limit was computed below 60%. At 95–99% completion, imperfect clearing still gives only 25.9% prevention. The extension includes a failed directional prediction.

Table T10: Leading-order limits (%) of the three primary profiles at three completion ranges, both workflows, with between-profile gaps (percentage points). Limits, not finite-p results; they do not determine absolute rates.

$h_0 \to h_1$	Workflow	Independent	Clustered	Imperfect	Ind.−clu.	Ind.−imp.	Clu.−imp.
0.80→0.90	Original	75.71	57.11	42.43	18.6	33.3	14.7
	Alternative	75.72	57.11	42.42	18.6	33.3	14.7
0.90→0.95	Original	75.13	53.65	27.99	21.5	47.1	25.7
	Alternative	75.18	53.65	27.99	21.5	47.2	25.7
0.95→0.99	Original	95.74	81.16	25.93	14.6	69.8	55.2
	Alternative	96.01	81.18	25.92	14.8	70.1	55.3

Raw and survival-weighted bridge counts. Raw candidate pairs are dominated by long bridges, yet survival weighting removes most of them (Table T11): 87.4% of raw carrier-pickup-to-susceptible-contact pairs span at least 13 opportunities, whereas bridges longer than two opportunities carry only 1.0%, 26.2% and 7.4% of the original-workflow baseline coefficient in the three primary profiles. The number of possible bridges alone therefore does not determine the leading-order rate.

Table T11: Unweighted carrier-pickup-to-observed-susceptible-contact pairs before a worker reset, original workflow. Pickups may originate in warmup; these frequencies ignore isolation weights and hand survival and are not survival-weighted transmission shares. Every odd count is zero: under the entry/exit rule a hand path between zones crosses an exit and an entry for each zone change.

Opportunities	Raw pairs	Raw fraction (%)
0	10	0.000
1	0	0.000
2	3158292	2.931
3	0	0.000
4	3130732	2.905
5	0	0.000
6	620685	0.576
7	0	0.000
8	2867327	2.661
9	0	0.000
10	2998258	2.782
11	0	0.000
12	782119	0.726
$\geq$13	94206242	87.419

Worker resets truncate bridges while retaining completion memory. Raw pairs ignore isolation weights and hand survival; weighted K includes both.

T4.3 The prespecified alternative-workflow coefficient study

Design. The protocol preceded this study's outcomes. The finite-p alternative-workflow results (Section T7) and the original-workflow limits were already known, so the coefficient values and the three predictions below are prospective, while any comparison with the finite-p results is post hoc. The study used the same 256 backgrounds and every scientific setting of the original coefficient study (3% admission carriage, clean hands and sites, $p = 0$, 90+56 days, care-zone opportunities, the frozen 30-bed model, the same profiles and h-pairs). The only change was the alternative workflow rules (Table T3), taken from the frozen alternative-workflow study code: vacated sites are cleared at discharge and relocation; a detected carrier moves only into an empty isolation bed; if none is empty it stays in place. At $p = 0$ site clearing cannot change any hand bridge because no site is ever contaminated, so it is retained only for fidelity; the isolation rule can change contact histories and isolation weights and therefore K.

Predictions written before outcomes. (1) The three primary limits change by less than 2 percentage points from 75.1%, 53.6% and 28.0%. (2) For .90$\rightarrow$.95 completion, $\Delta(e)$ is negative at $e = 1$ and positive at $e = .8$. (3) The crossing e^*, by linear interpolation in e between .9 and 1 at $\rho = .5$, lies within .05 of the isolated two-opportunity value .9235. All three were to be reported whatever the outcome.

Implementation and checks. An adapter combined the frozen bridge hooks with the alternative workflow. Archived original-mode records matched bitwise on 10 checked seeds; rule tests and two full-horizon lockstep checks passed. These were internal automated and delegated checks, not external peer review. All 256 records completed with no seed replacement. Before launch

the implementer saw three benchmark outcomes while testing the summary script; the protocol, settings and code then remained unchanged. The full 22-pair surface was selected by the declared compute-budget rule. Exact tests, review notes, runtimes and operational limitations are preserved in the execution records (Section T13).

Results. Alternative limits were 75.18%, 53.65% and 27.99% versus 75.13%, 53.65% and 27.99% on the same original-workflow seeds (Table T7). All 22 changes were below 0.3 percentage points; 21 were below 0.06. The largest (+0.27) was at .95→.99, $\rho = 0$, $e = 1$, where K_1 is smallest. All three predictions held (Table T12). Site clearing cannot act at $p = 0$; isolation changed every history and lowered both coefficients, so ratio invariance was not guaranteed. The predictions were low-risk: tolerance 2 points versus realized .05 for the primary limits, and .05 versus .005 for the historical two-point crossing statistic. Several paired intervals exclude zero, reflecting strong pairing; changes of .001–.3 points have no substantive interpretation.

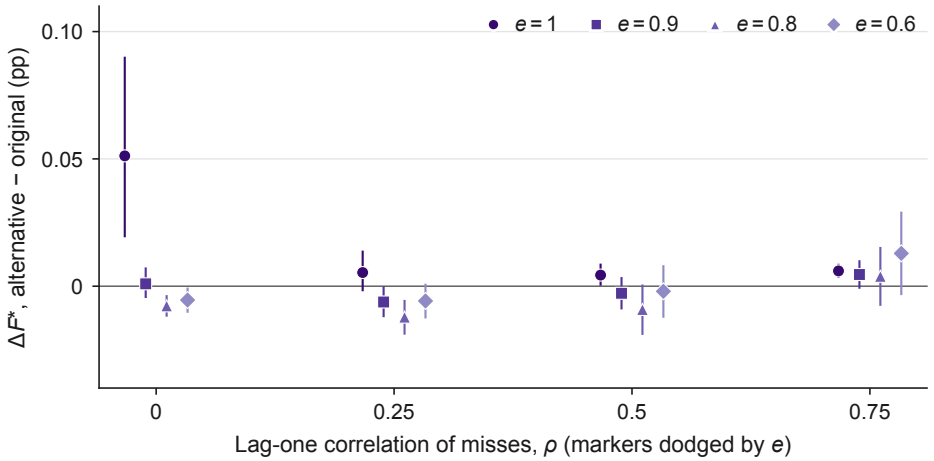


Fig. T1 Alternative minus original leading-order limit on the same 256 background seeds, completion .90→.95, with paired whole-seed bootstrap 95% intervals (20,000 draws; background Monte Carlo variation only); markers dodged by e. Every difference shown is below 0.06 percentage points; the largest of all 22 rows, +0.27, is at .95→.99, $\rho = 0$, $e = 1$.

Coefficient changes across workflows (post hoc). The alternative workflow lowered K_0 by 3.8–4.0% in all 22 rows (independent profile: −3.9% [−6.8, −0.8]) and K_1 by 3.8–4.0% in 21 rows. The exception was a 10.2% fall in K_1 at .95→.99, $\rho = 0$, $e = 1$, where prevention rose from 95.74% to 96.01%, near the isolated two-opportunity value 96.0%. Here K_1 is smallest, so the loss of zero-opportunity bridges matters most. Such bridges contributed at most 0.3% of original K; at $p = 0$ they are hand bridges, not environmental routes. None remained in the alternative workflow. The independent profile's per-seed K_1/K_0 standard deviation fell from .0034 (range .2481–.2786) to .00003. The survival-weighted opportunity mix otherwise changed little (Table T13).

No ordered bridge history matched across workflows (0 of 256): same backgrounds means same seeds, not same histories. Imported carriers averaged 9.20 versus 9.53 per observation window; patient-days averaged 1,640.4 versus 1,640.0. Alternative windows averaged 4.92 relocations, 1.10 detected-patient-days waiting in shared beds and no swaps. Per-worker independent K_0 changed by up to 2.6-fold, but role shares stayed close: nurses 86.7/86.1%, physicians 10.6/11.0% and radiography 2.7/2.9%. Both banks agreed within .002 percentage points in every primary limit in both

Table T12: The three prespecified predictions. Prediction 1 compares the alternative-workflow limit with the published rounded original value, as the protocol states; the paired difference from the exact same-seed original value and its whole-seed bootstrap 95% interval are also shown. Δ values are percentage points.

Prediction	Reference	Result	Detail	Verdict
1. Primary limit changes by < 2 pp: independent misses	75.1	75.18	+0.051 [+0.019, +0.090]	Held
1. Primary limit changes by < 2 pp: clustered misses	53.6	53.65	+0.004 [0.000, +0.009]	Held
1. Primary limit changes by < 2 pp: imperfect clearing	28.0	27.99	−0.008 [−0.012, −0.004]	Held
2. Sign reversal, $\Delta(1) = F^*(\rho{=}.5, 1) - F^*(0, 1)$	—	−21.48 / −21.53	original / alternative; all 20,000 draws < 0	Held
2. Sign reversal, $\Delta(.8) = F^*(\rho{=}.5, .8) - F^*(0, .8)$	—	+9.02 / +9.02	original / alternative; all 20,000 draws > 0	Held
3. Crossing e^* (protocol two-point statistic, $e = .9$ to 1) within .05 of .9235	.9235	0.9187 / 0.9187	distance 0.0048 / 0.0048	Held

The protocol reference .9235 is the exact root of the isolated two-opportunity crossing (.92353, Proposition T6). Prediction 3 used the protocol's two-point statistic, a linear interpolation between $e = .9$ and 1; its value is shown only as the record of this prediction. Because $\Delta(e)$ is strongly nonlinear in e, that statistic is not a location estimate: the fine-grid original-workflow crossing is .93361 (Section T4.4), also within .05 of .9235. No alternative-workflow fine grid was computed. The later estimate does not rescore these historical predictions.

workflows. Almost every surviving bridge still crossed two opportunities, so redistribution changed the ratio little. This explanation is post hoc. It does not establish invariance to bed management in general: the leading term cannot see ungated environmental chains, and the combined workflow change does not isolate either rule's effect. Differences in finite-p prevention therefore lie outside this leading term.

Table T13: Opportunity-count mix of the survival-weighted leading-order coefficient: percentage of mean K contributed by bridges with k intervening hygiene opportunities, primary profiles, same 256 backgrounds. Under the entry/exit rule every count between zones is even, so the columns are $k = 0$, $k = 2$ and $k \geq 4$ (odd counts contribute 0). A $k = 0$ bridge is a hand bridge with no intervening opportunity.

		Original workflow			Alternative workflow		
Profile	Arm	$k = 0$	$k = 2$	$k \geq 4$	$k = 0$	$k = 2$	$k \geq 4$
Independent misses	control (90%)	0.07	98.95	0.98	0.00	99.02	0.98
	policy (95%)	0.28	99.47	0.25	0.00	99.75	0.25
Clustered misses	control (90%)	0.01	73.83	26.16	0.00	73.93	26.07
	policy (95%)	0.02	76.03	23.95	0.00	76.14	23.86
Imperfect clearing	control (90%)	0.01	92.62	7.37	0.00	92.68	7.32
	policy (95%)	0.01	94.50	5.49	0.00	94.56	5.44

Finite-p comparison (post hoc). All alternative-workflow fractions were within 1.4 points of their limits and inside their 95% intervals (Table T14). In the original workflow, the independent fraction was 7.6 points below its limit, or 8.4 after common-baseline interpolation; the other profiles were within 2.3 points. Yet $1000p^2\overline{K}_h/D^{(0)}$ explains only 31–60% of alternative control rates (25–57% original), so higher-order absolute contributions are large. They changed the alternative control

and policy rates nearly proportionally; this is an observed property, not a guarantee. Every interval also contains the simpler two-opportunity comparator, so interval overlap cannot distinguish it from the full leading-order limit. Finite-p outcomes preceded this protocol.

Table T14: Post hoc, descriptive comparison of the alternative-workflow finite-p results (512 paired seeds per profile, once-selected p) with the alternative-workflow leading-order limit, and leading-order versus measured absolute rates (per 1,000 patient-days; ratio in parentheses).

Profile	Selected p	Finite-p F (%) [95%]	Limit (%)	$F-$limit (pp)	Control rate: leading / measured	Policy rate: leading / measured
Independent misses	0.16511	76.5 [74.1, 78.7]	75.18	+1.31	0.89 / 2.90 (0.31)	0.22 / 0.68 (0.32)
Clustered misses	0.08063	53.7 [49.7, 57.5]	53.65	+0.05	1.56 / 2.62 (0.59)	0.72 / 1.21 (0.60)
Imperfect clearing	0.07614	27.3 [22.2, 31.9]	27.99	−0.73	1.58 / 2.63 (0.60)	1.14 / 1.91 (0.60)

Interpretation. Agreement with finite-p fractions is descriptive, not validation: the fixed-horizon $p \to 0$ limit has no finite-p error bound. Neither agreement nor the combined workflow comparison identifies either rule's effect. Cross-platform reproducibility was checked only in original-workflow mode; alternative coefficient records ran on macOS (Section T13).

T4.4 Prespecified coefficient extension: crossing, policy definition and parameter grid

Design. The extension protocol specified 160 additional profile-pair definitions: 14 crossing rows, 10 completion-range rows, four transition-defined policy rows and a 11×12 parameter grid. Together with 22 archived pairs this gives 182 pairs on each of the same 256 original-workflow background seeds. Duplicate parameter definitions were retained as numerical checks. All backgrounds used the unchanged 30-bed model, 3% admission carriage, $p = 0$, the care-zone entry/exit rule and 90 warmup plus 56 observation days. Coefficient arms consume no random numbers and are fed the same event stream; these rows are not independent new ICU histories.

For every seed the ordered event hash, three primary profiles, all 22 archived pairs and duplicate pairs matched their archived references bitwise. All 256 records passed, none was excluded or replaced, and both benchmark seeds reproduced their new arms on rerun. The 10-worker local batch used 54.89 measured CPU-hours (55.03 whole-process CPU-hours), about 12% above the 48.87 projected by the 2-seed benchmark; the overrun, attributed to machine load, was reported and not acted on. The 60-hour budget governed only the plan decision, which used the projection. Disclosure: before launch, the implementer saw outcomes on the 2 benchmark seeds while testing the summary script; the plan, protocol and code did not change afterwards. Because coefficient Monte Carlo error is small, those outputs were nearly as informative as the full study (post hoc): over all 32,640 two-seed subsets of the 256 backgrounds, the standard deviation of the .70→.80, $e = .8$ clustering contrast was 0.02 points, and the two benchmark seeds gave +0.251 points against +0.254 from all 256. The protocol's precedence therefore rests on its file timestamp (01:57 EDT, before the benchmark runs began at 02:19) and on its hash, first recorded at code review (03:07); the protocol file was not under version control. The protocol, launch notes and reviewed code were preserved. The original and independently implemented summaries agreed on the four prediction decisions and numerical results. Point estimates are $F^* = 1 - \text{mean}(K_1)/\text{mean}(K_0)$; intervals use 20,000 synchronous whole-seed bootstrap draws, with the declared seed 20261023. They describe

background Monte Carlo uncertainty only. Prediction decisions use the prespecified point-estimate criteria, not multiplicity-adjusted tests.

Table T15: Four predictions declared before the coefficient extension, with criteria as written in the protocol. Failed predictions are retained. D is the clustering contrast in percentage points; the crossing interval includes background Monte Carlo variability only.

#	Criterion	Observed	Decision
1	ICU crossing e^* (root of $D(e) = F^*(\rho{=}.5) - F^*(\rho{=}0)$ at h .90→.95) lies in [.91, .935]	0.93361 [0.93358, 0.93365]; one sign change in all 20,000 draws	Held
2	At $e = .8$: $D < 0$ for .60→.70 and .70→.80, $D > 0$ for .80→.90; at $e = 1$, $D < 0$ in every range	.70→.80, $e = .8$: $D = +0.25$ pp, predicted < 0 (all 20,000 draws > 0); other 7 of 8 components as predicted	Failed
3	Transition-defined policy: $\lvert D(e = 1)\rvert < 5$ pp (against 21.5 pp per boundary)	+0.12 pp [0.12, 0.12]	Held
4	Renewal comparator: $\max \lvert F_{\text{geo}} - F^*\rvert \leq 3.5$ pp over the 132-row grid	4.32 pp ($\rho = 0.95$, $e = 0.50$)	Failed

Crossing and completion ranges. Prediction 1 held. All 20,000 draws had one crossing on the evaluated grid, between .93 and .94; linear interpolation gives .93361 [.93358,.93365]. This supersedes the earlier two-point location estimate, but interpolation error exceeds the bootstrap interval. Near-root second differences are −.30 and −.33 points on the .01 grid; concavity puts the linear root left of the smooth root. Post hoc quadratic, cubic and quartic fits give .93379–.93382, outside the interval. On the isolated bridge the same linear method misses the known root .92353 by 2.2×10^{-4} (estimate .92331), about three interval widths; a quadratic misses by less than 2×10^{-5}. Thus about .934 is appropriate, and both estimates satisfy Prediction 1.

Table T16 shows the nonlinear $\Delta(e)$, largest near .8. The crossing is above the isolated root and near the renewal value .936. Per-length roots (four opportunities: .845) do not bound a mixture's crossing: the two arms weight bridge lengths differently (Table T13).

Prediction 2 failed in one of its eight components. At $e = .8$, the clustering contrast for .70→.80 was +0.254 percentage points, whereas the protocol predicted $D < 0$ (isolated bridge −2.383); its background interval was [0.250, 0.258] points (Table T15 gives the bootstrap draw count). The .60→.70 contrast remained negative (−0.935 points), and the .80→.90 contrast was positive (+4.379 points). At perfect clearing the contrast remained negative at every tested range. The risk this prediction took lay mainly in the three $e = .8$ signs, of which two held: three of the five $e = 1$ components (.80→.90, .90→.95 and .95→.99) used only archived pairs whose outcomes were known before the protocol was written, and the two new $e = 1$ components had isolated values of −13.4 and −17.1 points. Post hoc and descriptively, in every range the ICU contrast exceeds the isolated two-opportunity value, by 0.66 to 5.29 points, so at $e = .8$ the full-coefficient reversal extends to lower completion than the isolated bridge suggests: the sign change lies between the .60→.70 and .70→.80 ranges, not between .70→.80 and .80→.90. The isolated formula therefore does not determine every full-ICU sign, even within the rare-transmission limit (Table T17).

Transition-defined policy. Direct-transition compliance of 90%→95% requires per-boundary .683772 → .776393 at $\rho = 0$ and .829180 → .908392 at $\rho = .5$ (unrounded roots in computation). Prediction 3 held: the perfect-clearing ICU contrast was 0.121 points, versus −21.5 under the

Table T16: Crossing scan: original-workflow F^* (%) at completion .90→.95, and $\Delta = F^*(.5, e) - F^*(0, e)$ (percentage points). Brackets are whole-seed bootstrap 95% intervals.

e	$F^*(0, e)$	$F^*(.5, e)$	Δ	95% interval
0.50	11.52	15.44	3.914	[3.911, 3.918]
0.60	14.93	20.54	5.608	[5.604, 5.613]
0.70	19.96	27.62	7.654	[7.649, 7.659]
0.75	23.46	32.04	8.572	[8.567, 8.577]
0.78	26.03	34.96	8.931	[8.926, 8.936]
0.80	27.99	37.01	9.018	[9.013, 9.023]
0.82	30.20	39.12	8.922	[8.917, 8.927]
0.85	34.07	42.35	8.275	[8.271, 8.280]
0.90	42.60	47.55	4.948	[4.944, 4.953]
0.91	44.73	48.51	3.774	[3.770, 3.779]
0.92	47.04	49.41	2.371	[2.366, 2.377]
0.93	49.55	50.26	0.708	[0.702, 0.714]
0.94	52.29	51.04	-1.250	[-1.257, -1.242]
0.95	55.28	51.74	-3.542	[-3.551, -3.532]
0.96	58.55	52.34	-6.208	[-6.219, -6.196]
0.98	66.08	53.24	-12.842	[-12.859, -12.822]
1.00	75.13	53.65	-21.485	[-21.513, -21.450]

Table T17: Completion-range contrasts in the original-workflow coefficient. Fractions are percentages; Δ and the isolated comparator are percentage points. All rows use the same 256 backgrounds. The isolated comparator has no Monte Carlo uncertainty.

$h_0 \to h_1$	e	$F^*(0, e)$	$F^*(.5, e)$	Δ	95% interval	Isolated Δ_2
0.60→0.70	0.80	33.85	32.92	-0.935	[-0.940, -0.931]	-3.518
0.60→0.70	1.00	47.45	36.80	-10.648	[-10.653, -10.643]	-13.393
0.70→0.80	0.80	37.14	37.39	0.254	[0.250, 0.258]	-2.383
0.70→0.80	1.00	57.65	43.16	-14.494	[-14.499, -14.490]	-17.094
0.80→0.90	0.80	42.43	46.81	4.379	[4.375, 4.383]	1.202
0.80→0.90	1.00	75.71	57.11	-18.601	[-18.608, -18.593]	-20.833
0.90→0.95	0.80	27.99	37.01	9.018	[9.013, 9.023]	5.559
0.90→0.95	1.00	75.13	53.65	-21.485	[-21.513, -21.450]	-22.727
0.95→0.99	0.80	25.93	42.55	16.620	[16.611, 16.629]	11.331
0.95→0.99	1.00	95.74	81.16	-14.579	[-14.732, -14.396]	-15.238

per-boundary policy (Table T18). This was low-risk: its 5-point tolerance was about 40 times the observed contrast. At $e = 1$, matching transition compliance matches q_2, which carries ρ in Proposition T5; the isolated contrast is exactly zero. The small ICU departure is new information. At $e = .8$ the ICU contrast remained 7.446 points, so matching transition compliance does not remove clustering dependence with imperfect clearing.

The policies differ because their per-boundary completion levels differ by profile. Forecasts depend on the chosen score; neither score is an artifact. Boundary electronic monitoring and WHO direct observation measure different units (Section T2). This comparison assumes stationary Markov completion and one exit and entry per direct move; it does not validate clinical audit practice.

Table T18: Transition-defined 90%→95% policy: original-workflow F^* (%), contrasting $\rho = .5$ with $\rho = 0$. Each uses its own per-boundary completion levels. Δ and its 95% interval are percentage points.

e	$F^*(0, e)$	$F^*(.5, e)$	Δ	95% interval	Isolated Δ_2
0.80	34.18	41.62	7.446	[7.444, 7.447]	6.105
1.00	52.36	52.48	0.121	[0.119, 0.124]	0.000

Expanded renewal comparison. Prediction 4 failed. On the 132-node grid the maximum absolute discrepancy was 4.32 percentage points, exceeding the predicted 3.5-point ceiling, at $\rho = .95, e = .5$. Post hoc and descriptively, the renewal comparator exceeded F^* at every node; 49 nodes were within 0.5 points and 81 within 1 point; errors grow toward high ρ and low e (3.91 points at $\rho = .9$, $e = .5$); and for $\rho \leq .7$, a cut chosen after the failure was seen, the maximum was 2.48 points, at $\rho = .7$, $e = .6$ (the grid has no $\rho = .75$ row). The historical 22-row maximum of 2.78 points remains correct on its smaller domain. The complete expanded grid is given in Table T19. These coefficients also supply the interpolated measurement projection in Section T11; no new per-opportunity histogram was recorded for these non-primary arms. None of the extension tests the sign reversal at finite p or supplies a finite-p error bound.

Table T19: Expanded original-workflow coefficient grid: completion .90→.95, 256 background seeds. F^* and F_{geo} are percentages; error is $F_{\text{geo}} - F^*$ in percentage points. Intervals include background uncertainty only.

ρ	e	F^*	95% interval	F_{geo}	Error
0.00	0.50	11.52	[11.52, 11.53]	12.26	0.744
0.00	0.60	14.93	[14.92, 14.93]	15.48	0.554
0.00	0.70	19.96	[19.96, 19.97]	20.30	0.339
0.00	0.75	23.46	[23.46, 23.47]	23.70	0.242
0.00	0.78	26.03	[26.03, 26.03]	26.22	0.190
0.00	0.80	27.99	[27.99, 28.00]	28.15	0.158
0.00	0.82	30.20	[30.20, 30.20]	30.33	0.130
0.00	0.85	34.07	[34.07, 34.08]	34.17	0.094
0.00	0.90	42.60	[42.60, 42.61]	42.66	0.052
0.00	0.95	55.28	[55.26, 55.29]	55.31	0.034
0.00	0.98	66.08	[66.06, 66.10]	66.12	0.040
0.00	1.00	75.13	[75.09, 75.16]	75.19	0.056
0.10	0.50	12.05	[12.04, 12.05]	12.86	0.814
0.10	0.60	15.75	[15.75, 15.76]	16.39	0.636
0.10	0.70	21.24	[21.24, 21.24]	21.66	0.424

ρ	e	F^*	95% interval	F_{geo}	Error
0.10	0.75	25.02	[25.02, 25.03]	25.34	0.322
0.10	0.78	27.76	[27.76, 27.76]	28.03	0.266
0.10	0.80	29.83	[29.82, 29.83]	30.06	0.231
0.10	0.82	32.11	[32.10, 32.11]	32.31	0.199
0.10	0.85	36.00	[36.00, 36.01]	36.16	0.157
0.10	0.90	43.93	[43.92, 43.94]	44.03	0.101
0.10	0.95	53.55	[53.54, 53.56]	53.61	0.065
0.10	0.98	59.34	[59.33, 59.35]	59.39	0.054
0.10	1.00	62.38	[62.36, 62.39]	62.43	0.051
0.20	0.50	12.66	[12.66, 12.67]	13.57	0.913
0.20	0.60	16.68	[16.67, 16.68]	17.44	0.757
0.20	0.70	22.60	[22.60, 22.61]	23.16	0.554
0.20	0.75	26.62	[26.62, 26.63]	27.07	0.452
0.20	0.78	29.48	[29.48, 29.49]	29.88	0.393
0.20	0.80	31.61	[31.60, 31.61]	31.96	0.355
0.20	0.82	33.92	[33.91, 33.92]	34.24	0.320
0.20	0.85	37.74	[37.74, 37.75]	38.01	0.269
0.20	0.90	45.03	[45.02, 45.03]	45.22	0.196
0.20	0.95	52.69	[52.69, 52.70]	52.83	0.139
0.20	0.98	56.46	[56.45, 56.47]	56.57	0.115
0.20	1.00	57.98	[57.97, 57.99]	58.09	0.105
0.30	0.50	13.39	[13.39, 13.40]	14.44	1.050
0.30	0.60	17.74	[17.74, 17.75]	18.67	0.930
0.30	0.70	24.08	[24.08, 24.09]	24.83	0.749
0.30	0.75	28.29	[28.29, 28.30]	28.94	0.648
0.30	0.78	31.23	[31.22, 31.23]	31.81	0.586
0.30	0.80	33.37	[33.36, 33.38]	33.91	0.544
0.30	0.82	35.66	[35.65, 35.66]	36.16	0.503
0.30	0.85	39.35	[39.34, 39.35]	39.79	0.441
0.30	0.90	45.96	[45.95, 45.96]	46.30	0.340
0.30	0.95	52.20	[52.20, 52.21]	52.45	0.250
0.30	0.98	54.88	[54.87, 54.89]	55.09	0.212
0.30	1.00	55.79	[55.78, 55.80]	55.99	0.197
0.40	0.50	14.29	[14.28, 14.29]	15.53	1.242
0.40	0.60	19.00	[18.99, 19.00]	20.17	1.174
0.40	0.70	25.73	[25.72, 25.73]	26.75	1.023
0.40	0.75	30.07	[30.07, 30.08]	31.00	0.920
0.40	0.78	33.03	[33.03, 33.04]	33.89	0.851
0.40	0.80	35.15	[35.15, 35.16]	35.95	0.801
0.40	0.82	37.37	[37.37, 37.38]	38.12	0.748
0.40	0.85	40.86	[40.86, 40.87]	41.52	0.664
0.40	0.90	46.78	[46.78, 46.79]	47.30	0.515
0.40	0.95	51.90	[51.90, 51.91]	52.28	0.378
0.40	0.98	53.89	[53.89, 53.90]	54.22	0.322
0.40	1.00	54.49	[54.48, 54.49]	54.79	0.304
0.50	0.50	15.44	[15.43, 15.44]	16.94	1.508
0.50	0.60	20.54	[20.53, 20.54]	22.04	1.507
0.50	0.70	27.62	[27.61, 27.63]	29.00	1.381
0.50	0.75	32.04	[32.03, 32.04]	33.30	1.265
0.50	0.78	34.96	[34.95, 34.97]	36.14	1.175
0.50	0.80	37.01	[37.00, 37.02]	38.12	1.107
0.50	0.82	39.12	[39.11, 39.13]	40.15	1.032
0.50	0.85	42.35	[42.34, 42.35]	43.26	0.910
0.50	0.90	47.55	[47.55, 47.56]	48.24	0.690
0.50	0.95	51.74	[51.73, 51.74]	52.23	0.493
0.50	0.98	53.24	[53.24, 53.24]	53.66	0.417
0.50	1.00	53.65	[53.64, 53.65]	54.04	0.396

ρ	e	F^*	95% interval	F_{geo}	Error
0.60	0.50	16.99	[16.98, 17.00]	18.87	1.875
0.60	0.60	22.52	[22.51, 22.53]	24.46	1.943
0.60	0.70	29.89	[29.88, 29.90]	31.70	1.812
0.60	0.75	34.28	[34.27, 34.29]	35.93	1.651
0.60	0.78	37.10	[37.09, 37.11]	38.62	1.522
0.60	0.80	39.03	[39.02, 39.04]	40.46	1.422
0.60	0.82	40.98	[40.97, 40.99]	42.29	1.313
0.60	0.85	43.87	[43.87, 43.88]	45.01	1.134
0.60	0.90	48.33	[48.32, 48.33]	49.15	0.824
0.60	0.95	51.68	[51.68, 51.68]	52.24	0.564
0.60	0.98	52.81	[52.81, 52.82]	53.29	0.471
0.60	1.00	53.10	[53.09, 53.10]	53.55	0.449
0.70	0.50	19.26	[19.25, 19.27]	21.64	2.378
0.70	0.60	25.25	[25.23, 25.26]	27.73	2.483
0.70	0.70	32.76	[32.75, 32.77]	35.03	2.267
0.70	0.75	36.99	[36.98, 37.00]	39.00	2.016
0.70	0.78	39.60	[39.59, 39.61]	41.42	1.821
0.70	0.80	41.33	[41.32, 41.34]	43.01	1.676
0.70	0.82	43.05	[43.04, 43.06]	44.57	1.521
0.70	0.85	45.53	[45.52, 45.53]	46.80	1.278
0.70	0.90	49.16	[49.15, 49.16]	50.04	0.881
0.70	0.95	51.73	[51.73, 51.74]	52.31	0.573
0.70	0.98	52.56	[52.56, 52.56]	53.03	0.471
0.70	1.00	52.75	[52.75, 52.75]	53.20	0.448
0.80	0.50	22.95	[22.93, 22.96]	26.01	3.061
0.80	0.60	29.33	[29.31, 29.34]	32.41	3.077
0.80	0.70	36.63	[36.62, 36.65]	39.26	2.626
0.80	0.75	40.42	[40.41, 40.43]	42.65	2.233
0.80	0.78	42.64	[42.63, 42.65]	44.60	1.958
0.80	0.80	44.08	[44.07, 44.08]	45.84	1.765
0.80	0.82	45.45	[45.45, 45.46]	47.02	1.569
0.80	0.85	47.38	[47.38, 47.39]	48.66	1.276
0.80	0.90	50.07	[50.07, 50.08]	50.91	0.835
0.80	0.95	51.88	[51.87, 51.88]	52.40	0.521
0.80	0.98	52.43	[52.42, 52.43]	52.85	0.423
0.80	1.00	52.55	[52.55, 52.55]	52.95	0.402
0.90	0.50	30.00	[29.99, 30.02]	33.91	3.910
0.90	0.60	36.20	[36.18, 36.21]	39.69	3.489
0.90	0.70	42.24	[42.23, 42.25]	44.85	2.615
0.90	0.75	45.00	[44.99, 45.01]	47.09	2.089
0.90	0.78	46.51	[46.51, 46.52]	48.29	1.772
0.90	0.80	47.45	[47.45, 47.46]	49.02	1.566
0.90	0.82	48.33	[48.32, 48.33]	49.69	1.367
0.90	0.85	49.50	[49.50, 49.51]	50.60	1.091
0.90	0.90	51.06	[51.06, 51.06]	51.77	0.710
0.90	0.95	52.04	[52.04, 52.05]	52.51	0.463
0.90	0.98	52.33	[52.33, 52.33]	52.72	0.390
0.90	1.00	52.39	[52.39, 52.39]	52.77	0.375
0.95	0.50	36.57	[36.55, 36.59]	40.88	4.315
0.95	0.60	41.66	[41.65, 41.67]	45.10	3.443
0.95	0.70	46.01	[46.00, 46.02]	48.40	2.387
0.95	0.75	47.82	[47.81, 47.83]	49.70	1.877
0.95	0.78	48.77	[48.76, 48.78]	50.36	1.594
0.95	0.80	49.34	[49.34, 49.35]	50.76	1.419
0.95	0.82	49.87	[49.86, 49.88]	51.13	1.256
0.95	0.85	50.56	[50.56, 50.57]	51.60	1.038
0.95	0.90	51.45	[51.45, 51.45]	52.20	0.751

ρ	e	F^*	95% interval	F_{geo}	Error
0.95	0.95	51.99	[51.99, 52.00]	52.57	0.574
0.95	0.98	52.15	[52.15, 52.15]	52.67	0.522
0.95	1.00	52.18	[52.18, 52.19]	52.70	0.512

T5 Calibration, selection and baseline-matching checks

T5.1 Selection procedures

Every study selected the transmission probability p from baseline (control) acquisitions only, then held it fixed for the paired policy comparison; policy outcomes never selected p.

- *Antecedent exploration*: 27 conditions with 64 paired seeds each (1,728 records). The rule took the lowest exact target hit, otherwise the first increasing adjacent bracket, interpolating p linearly against pooled baseline rate, with no extrapolation or fallback. It selected $p =$ .07602158 (clustered misses) and .07483996 (imperfect clearing); its independent grid had no point selection.
- *Independent-profile calibration grid*: 128 seeds at $p =$ .13496571, .15 and .165 gave baseline rates 2.311, 3.558 and 5.027 per 1,000 patient-days. The declared rule (exact hit, otherwise lowest increasing bracket with linear interpolation of p against rate; fixed $p = .15$ fallback recorded as a failure) selected $p = .14327559$ from the first bracket. The calibration validity check passed.
- *Original holdout*: the three probabilities were frozen and evaluated on 512 fresh paired seeds per profile.
- *Opportunity-rule follow-up* and *alternative-workflow study* used their own declared rules (Sections T10 and T7.2). The alternative study interpolated on the log–log scale; the original rules were linear in p versus rate (or versus acquisition count in earlier refinement and validation rules). None of the original rules was logarithmic.

T5.2 All baseline-matching and target-band checks

The original holdout study declared six simultaneous 99.167% interval checks: each profile's baseline rate interval had to lie wholly inside [2.7, 3.3] per 1,000 patient-days, and each pairwise baseline-rate-ratio interval wholly inside $[1/1.1, 1.1]$. These are declared numerical tolerances, not tests of exact equality; any failed check prevents a matched-baseline claim, and passing depends on both precision and the underlying rates. All six failed, although the calibration validity check passed. The opportunity-rule follow-up added two target checks (entire 99.167% interval inside [2.7, 3.3]); both failed. These are the eight failed checks. The alternative-workflow study used descriptive 95% band checks; the clustered and imperfect checks failed. No refitting or sample extension was made in response to any failure. The coarse inherited-clock sweep also found no sampled match (Section T10).

Table T20: Every baseline-matching and target-band check, with verbatim estimates and intervals (rates per 1,000 patient-days).

#	Study	Check	Estimate	Interval	Allowed band	Outcome
Original holdout: six simultaneous 99.167% interval checks (200,000 whole-seed draws)						
1	Original holdout	Independent baseline rate	2.829	[2.596, 3.069]	[2.7, 3.3]	Not met
2	Original holdout	Clustered baseline rate	2.686	[2.489, 2.887]	[2.7, 3.3]	Not met
3	Original holdout	Imperfect baseline rate	2.801	[2.609, 2.998]	[2.7, 3.3]	Not met
4	Original holdout	Independent/clustered rate ratio	1.0532	[0.9518, 1.1639]	[0.9091, 1.1000]	Not met
5	Original holdout	Independent/imperfect rate ratio	1.0097	[0.9092, 1.1181]	[0.9091, 1.1000]	Not met
6	Original holdout	Clustered/imperfect rate ratio	0.9587	[0.8734, 1.0520]	[0.9091, 1.1000]	Not met
Opportunity-rule follow-up: selected inherited-clock target checks (99.167% family level)						
7	Clock follow-up	Clustered baseline rate	3.09	[2.85, 3.33]	[2.7, 3.3]	Not met
8	Clock follow-up	Imperfect baseline rate	2.85	[2.62, 3.09]	[2.7, 3.3]	Not met

Alternative-workflow study: descriptive 95% target-band checks, band [2.7, 3.3]			
Profile	Baseline rate	95% interval	Interval in band?
Independent	2.904	[2.756, 3.052]	inside
Clustered misses	2.621	[2.480, 2.765]	not inside
Imperfect clearing	2.626	[2.484, 2.769]	not inside

T5.3 Selection uncertainty

Retrospective bootstraps of the saved calibration grids describe how variable the selected p would have been. They condition on the already chosen grids and exclude uncertainty from grid choice and interpolation error; they were not propagated into the policy intervals (Section T5.5 gives an approximate propagation).

Table T21: Independent-profile selection uncertainty from the three-probability baseline grid (50,000 draws, seed 20261010). Conditional intervals retain only exact-hit/increasing-bracket selections; fallback-inclusive intervals apply the fixed $p = .15$ fallback and retain failures. Selection diagnostics, not propagated holdout uncertainty.

Cohort	Seeds	Selected p [conditional 95%]	p [fallback-inclusive 95%]	Failed draws	Failure (%)	Nonmonotone
pooled	128	0.14328 [0.14004, 0.14792]	0.14328 [0.14004, 0.14792]	0 / 50000	0.00%	0
A	64	0.14301 [0.13739, 0.15084]	0.14301 [0.13741, 0.15083]	740 / 50000	1.48%	200
B	64	0.14343 [0.13967, 0.14984]	0.14343 [0.13967, 0.14984]	0 / 50000	0.00%	112

Table T22: Retrospective uncertainty in the antecedent 27-condition exploratory selection (50,000 joint whole-seed draws, seed 20261011). Each pooled analysis has 64 seeds; historical local and cluster subsets have 32 each, and the stratified diagnostic resamples 32 from each. Intervals condition on defined selections; any failed draw makes the unconditional interval unavailable.

Profile	Cohort	Point p	Conditional 95% interval	Failed draws	Unconditional
Independent	pooled64	undefined	[0.13179, 0.13496]	49,522	unavailable
Clustered misses	pooled64	0.07602	[0.07346, 0.07873]	0	as conditional
Imperfect clearing	pooled64	0.07484	[0.07243, 0.07785]	0	as conditional
Independent	local32	undefined	[0.13181, 0.13484]	49,721	unavailable

Profile	Cohort	Point p	Conditional 95% interval	Failed draws	Unconditional
Clustered misses	local32	0.07693	[0.07342, 0.08058]	7	unavailable
Imperfect clearing	local32	0.07556	[0.07206, 0.08087]	358	unavailable
Independent	cluster32	undefined	[0.12829, 0.13485]	44,158	unavailable
Clustered misses	cluster32	0.07497	[0.07097, 0.07942]	13	unavailable
Imperfect clearing	cluster32	0.07418	[0.07072, 0.07789]	3	unavailable
Independent	32+32	undefined	[0.13196, 0.13496]	49,491	unavailable
Clustered misses	32+32	0.07602	[0.07350, 0.07869]	0	as conditional
Imperfect clearing	32+32	0.07484	[0.07239, 0.07785]	1	unavailable

The bank-A independent conditional interval uses 49,260 valid draws; its 740 failed draws remain visible. The old independent exploratory grid has no point selection despite occasional defined bootstrap draws. The original 20,000-draw exploratory result was reconstructed before extension to 50,000 draws, and all 1,536 holdout and 1,728 exploratory record hashes were verified unchanged. No conclusion about the cause of the baseline undershoot follows from linear interpolation alone.

T5.4 Baseline-mismatch elasticity and implied bias

Baseline-mismatch estimator. Seed-paired original-workflow conditions varying p estimate $\beta_B = \mathrm{d}F/\mathrm{d}\ln B$, in percentage points per natural-log baseline (Table T23). A 10% baseline increase changes F by $\ln(1.1)\beta_B \approx 0.095\beta_B$; estimated bias relative to 3.0 is $F(B_{\mathrm{obs}}) - F(3.0) \approx \beta_B \ln(B_{\mathrm{obs}}/3)$. Positive bias means observed F exceeds its estimated value at 3.0.

Intervals use 20,000 whole-seed percentile draws (seed 20261021), preserving shared seeds within banks. Four 128-seed holdout strata preserve overlap with the independent upper bank. Recomputed intervals differ from the archived 200,000-draw values by at most .12 points; point values agree. Independent numerical verification and rounding differences are documented in the repository execution records.

Table T23: Original-workflow conditions used for the slopes (care-zone opportunities, completion .90 vs .95). Control rate per 1,000 patient-days and F (%) with 95% intervals. Conditions in the same bank share seeds.

Condition	Seed bank	Seeds	p	Control rate B [95%]	F (%) [95%]
Clustered, 0.75×	Sensitivity bank (A/B 128)	256	0.057016	1.296 [1.172, 1.422]	60.48 [53.77, 66.42]
Clustered, center	Sensitivity bank (A/B 128)	256	0.076022	2.630 [2.416, 2.849]	55.35 [49.91, 60.37]
Clustered, 1.25×	Sensitivity bank (A/B 128)	256	0.095027	5.029 [4.715, 5.356]	53.58 [48.88, 57.87]
Imperfect, 0.75×	Sensitivity bank (A/B 128)	256	0.056130	1.253 [1.134, 1.372]	28.90 [18.19, 38.47]
Imperfect, center	Sensitivity bank (A/B 128)	256	0.074840	2.809 [2.599, 3.027]	29.77 [22.19, 36.83]
Imperfect, 1.25×	Sensitivity bank (A/B 128)	256	0.093550	5.928 [5.559, 6.299]	32.69 [27.43, 37.58]
Independent, center (matched)	Holdout seeds A[:128]+B[:128]	256	0.143276	2.777 [2.525, 3.039]	66.29 [60.62, 71.21]
Independent, 1.25× (matched)	Holdout seeds A[:128]+B[:128]	256	0.179094	7.289 [6.746, 7.849]	71.01 [67.22, 74.42]
Independent, environmental bank	Environmental bank (A/B 256)	512	0.080979	0.395 [0.347, 0.444]	67.77 [60.38, 74.06]

Condition	Seed bank	Seeds	p	Control rate B [95%]	F (%) [95%]
Independent, environmental bank	Environmental bank (A/B 256)	512	0.107973	1.000 [0.916, 1.089]	63.93 [58.64, 68.76]
Independent, environmental bank	Environmental bank (A/B 256)	512	0.134966	2.288 [2.134, 2.444]	63.51 [59.41, 67.36]

Table T24: Slopes. $\beta_B = \mathrm{d}F/\mathrm{d}\ln B$ and $\mathrm{d}F/\mathrm{d}\ln p$ in percentage points; $\gamma = \mathrm{d}\ln B/\mathrm{d}\ln p$. Two-point slopes are paired finite differences; OLS slopes are equal-weight three-point fits. Every β_B interval includes zero.

Profile	Segment	Baselines	β_B [95%]	Per +10% baseline [95%]	$\mathrm{d}F/\mathrm{d}\ln p$ [95%]	γ [95%]
Independent	Center→1.25× (primary)	2.78–7.29	4.89 [−1.94, 11.19]	0.47 [−0.19, 1.07]	21.15 [−7.85, 50.93]	4.32 [3.84, 4.82]
	Env. bank .081→.108	0.40–1.00	−4.14 [−14.21, 4.91]	−0.39 [−1.35, 0.47]	−13.36 [−42.24, 16.93]	3.23 [2.73, 3.74]
	Env. bank .108→.135	1.00–2.29	−0.51 [−9.05, 7.00]	−0.05 [−0.86, 0.67]	−1.88 [−31.02, 27.92]	3.71 [3.22, 4.20]
	Env. bank, OLS	0.40–1.00–2.29	−2.46 [−7.14, 2.39]	−0.23 [−0.68, 0.23]	−8.58 [−24.14, 8.48]	3.43 [3.17, 3.70]
Clustered	Center→1.25× (primary)	2.63–5.03	−2.72 [−14.98, 7.91]	−0.26 [−1.43, 0.75]	−7.90 [−40.03, 24.59]	2.91 [2.50, 3.33]
	0.75× →center	1.30–2.63	−7.25 [−19.64, 3.62]	−0.69 [−1.87, 0.35]	−17.83 [−43.97, 9.65]	2.46 [2.09, 2.84]
	Three-point OLS	1.30–2.63–5.03	−5.12 [−11.22, 0.93]	−0.49 [−1.07, 0.09]	−13.70 [−28.94, 2.36]	2.65 [2.44, 2.85]
Imperfect	Center→1.25× (primary)	2.81–5.93	3.90 [−8.60, 15.13]	0.37 [−0.82, 1.44]	13.06 [−26.82, 53.73]	3.35 [2.98, 3.71]
	0.75× →center	1.25–2.81	1.09 [−15.68, 15.86]	0.10 [−1.49, 1.51]	3.06 [−39.88, 48.60]	2.81 [2.42, 3.21]
	Three-point OLS	1.25–2.81–5.93	2.42 [−4.92, 9.99]	0.23 [−0.47, 0.95]	7.22 [−15.24, 31.59]	3.03 [2.83, 3.25]

The center-to-1.25×center segment is the primary slope because it is the only segment whose baselines bracket 3.0 for every profile. The independent primary segment spans a 2.6-fold baseline range (2.78 to 7.29), and the environmental full-model segments below the target have the opposite point sign, so local linearity is weakest for that profile, although its observed shortfalls are the smallest. The independent primary slope equals the upper-minus-matched-center contrast of the opportunity-rule study (+4.7 percentage points, Table T47) divided by its log-baseline span. Applying the primary slopes from the center to 3.0 reproduces the common-baseline F_{cb} of Section T6 to within 0.027 percentage points.

Two local estimators are reported. *Piecewise* uses the low-to-center slope for any part of the span below the sensitivity center and the center-to-high slope above it; *primary* uses the center-to-high slope throughout. Their point estimates coincide when the observed baseline lies above the sensitivity center. Joint intervals resample the slope and the observed baseline together.

Table T25: Implied bias $F(B_{\mathrm{obs}}) - F(3.0)$ in percentage points with joint 95% intervals, the range of point estimates across every slope estimator in Table T24, and F adjusted to 3.0 with the piecewise estimator. Alternative-workflow rows reuse original-workflow slopes (a transfer assumption).

Workflow	Profile	B_{obs}	vs 3.0	Bias, piecewise [95%]	Bias, primary [95%]	Range, all estimators	F obs. (%)	F at 3.0 (%) [95%]
Original	Independent misses	2.829	−5.7%	−0.29 [−1.08, 0.07]	−0.29 [−1.08, 0.07]	−0.29 to +0.24	67.6	67.9 [64.8, 70.9]
	Clustered misses	2.686	−10.5%	+0.30 [−0.83, 1.55]	+0.30 [−0.89, 1.78]	+0.30 to +0.80	55.9	55.6 [51.5, 59.3]
	Imperfect clearing	2.801	−6.6%	−0.26 [−1.21, 0.39]	−0.27 [−1.25, 0.63]	−0.27 to −0.07	27.0	27.2 [21.9, 32.2]
Alternative	Independent misses	2.904	−3.2%	−0.16 [−0.61, 0.12]	−0.16 [−0.61, 0.12]	−0.16 to +0.14	76.5	76.7 [74.4, 78.8]
	Clustered misses	2.621	−12.6%	+0.38 [−0.96, 1.77]	+0.37 [−1.09, 2.13]	+0.37 to +0.98	53.7	53.3 [48.9, 57.3]
	Imperfect clearing	2.626	−12.5%	−0.33 [−1.92, 0.76]	−0.52 [−2.13, 1.16]	−0.52 to −0.14	27.3	27.6 [22.5, 32.6]

Results and assumptions. Estimated bias is at most .52 points under the two local estimators (joint 95% intervals within $[-2.13, 2.13]$), or .98 using any tested slope. Estimated contrast bias is at most .89 points (intervals within $[-2.34, 3.27]$); all six adjusted contrasts still exclude zero at 98.333%. For scale, Monte Carlo half-widths are 2.30–5.24 points and declared contrasts 11.7–49.2 points. Every slope interval includes zero, so neither direction nor curvature is resolved.

Table T26: Implied bias in the prespecified contrasts (percentage points). Observed-contrast intervals are recomputed here so they can be combined draw by draw and differ slightly from the archived intervals (different stream and stratification); the archived intervals remain authoritative. Adjusted contrasts are post hoc and descriptive.

Workflow	Contrast	Observed [98.333%]	Bias, piecewise [95%]	Bias, primary [95%]	Adjusted to 3.0 [98.333%]
Original	Independent − clustered	11.7 [5.4, 17.8]	−0.59 [−2.09, 0.62]	−0.59 [−2.30, 0.67]	12.3 [6.3, 18.6]
	Independent − imperfect	40.6 [33.1, 48.5]	−0.03 [−1.01, 1.00]	−0.02 [−1.19, 1.02]	40.7 [33.5, 48.2]
	Clustered − imperfect	28.9 [21.2, 36.9]	+0.56 [−0.73, 2.14]	+0.57 [−0.92, 2.32]	28.4 [20.5, 36.4]
Alternative	Independent − clustered	22.8 [17.3, 28.6]	−0.54 [−1.99, 0.84]	−0.53 [−2.34, 0.96]	23.3 [17.6, 29.6]
	Independent − imperfect	49.2 [42.7, 56.0]	+0.17 [−1.00, 1.79]	+0.36 [−1.38, 2.02]	49.1 [42.4, 55.9]
	Clustered − imperfect	26.4 [18.7, 34.3]	+0.71 [−0.88, 2.85]	+0.89 [−1.32, 3.27]	25.7 [17.4, 33.7]

These post hoc adjustments assume local linearity in $\ln B$ and that changing p represents the relevant baseline correction. Seed, workflow or admission-mix differences need not have that response. Alternative-workflow results transfer original slopes because no alternative sensitivity bank exists; its grid checks only γ. Separate environmental-bank points (baselines .40–2.29; 512 seeds) supply secondary slopes only. Intervals omit interpolation-model, structural and clinical-input uncertainty; the estimates are not error bounds and cannot turn failed checks into passes. Adjusted 512-seed holdout fractions differ from 256-seed common-baseline estimates because their center samples differ (66.3/55.3/29.8% versus 67.6/55.9/27.0% at the same p).

T5.5 Propagating selection uncertainty

Each workflow selected p once. The implied shift in F from moving p to an end of its conditional 95% selection interval is $\mathrm{d}F/\mathrm{d}\ln p \times \ln(p_{\text{bound}}/p_{\text{selected}})$, using the primary slope (for the alternative workflow, the original-workflow slope). "With slope uncertainty" is a per-draw envelope: for each bootstrap draw of the slope the smaller and larger of the two endpoint shifts are taken, and the table reports the 2.5th percentile of the smaller and the 97.5th percentile of the larger. The Monte Carlo half-width is half the width of the archived, unrounded 95% interval of F. The combined half-width $\sqrt{\mathrm{MC}^2 + S^2}$ uses S, the largest absolute shift over the slope's point value and 95% endpoints at the two ends of the p interval; it treats the two sources as independent and symmetric and is descriptive only.

Selection uncertainty moves F by at most 0.67 percentage points at the point slope, or 2.12 points allowing for slope uncertainty; slope uncertainty, not the width of the p interval, dominates the wider bound. Imperfect clearing in the alternative workflow chose the lower bracket in 2,861 of 20,000 selection draws, and its archived interval includes those draws. As a sensitivity check for the transfer assumption, the baseline-on-p part can be replaced by the alternative grid's own log–log slope over the m1–m2 bracket (Table T28); the point shifts stay below 0.5 percentage points.

T6 Common-baseline post hoc interpolation

Design. After all eight matching checks and the bracketing outcomes were known, a secondary analysis of existing original-workflow records was declared; its protocol preceded the new bootstrap calculations. For each profile the lower condition is its fixed center probability and the upper condition is 1.25 times that probability. The independent comparison uses the 256 matched holdout/upper seeds; clustered misses and imperfect clearing use the 256 sensitivity seeds. No

Table T27: Selection uncertainty in p propagated to F (percentage points). Original selected-p intervals: independent from Table T21 (pooled row); clustered and imperfect from Table T22 (pooled64 rows, older 64-seed design). Alternative intervals: Table T30. All have 0 failed draws.

Workflow	Profile	Selected p [95%]	Interval vs p	$\mathrm{d}F/\mathrm{d}\ln p$ [95%]	Shift, point slope	Shift, slope envelope	MC half-width	Combined
Original	Independent	0.14328 [0.14004, 0.14792]	−2.3%/+3.2%	21.2 [−7.8, 50.9]	−0.48 to +0.67	−1.16 to +1.62	3.35	3.73
	Clustered	0.07602 [0.07346, 0.07873]	−3.4%/+3.6%	−7.9 [−40.0, 24.6]	−0.28 to +0.27	−1.42 to +1.39	3.70	3.95
	Imperfect	0.07484 [0.07243, 0.07785]	−3.2%/+4.0%	13.1 [−26.8, 53.7]	−0.43 to +0.52	−1.78 to +2.12	5.24	5.66
Alternative	Independent	0.16511 [0.15999, 0.17033]	−3.1%/+3.2%	21.2 [−7.8, 50.9]	−0.67 to +0.66	−1.60 to +1.58	2.30	2.80
	Clustered	0.08063 [0.07801, 0.08332]	−3.3%/+3.3%	−7.9 [−40.0, 24.6]	−0.26 to +0.26	−1.33 to +1.34	3.88	4.10
	Imperfect	0.07614 [0.07338, 0.07862]	−3.6%/+3.2%	13.1 [−26.8, 53.7]	−0.48 to +0.42	−1.99 to +1.74	4.85	5.24

Table T28: Alternative-workflow sensitivity: original-workflow β_B combined with the alternative grid's own γ over the m1–m2 bracket.

Profile	Alternative grid B, m1 to m2	Alternative γ [95%]	Original γ	$\mathrm{d}F/\mathrm{d}\ln p$ [95%]	Shift, point slope
Independent misses	1.92 to 17.03	3.15 [2.96, 3.35]	4.32	15.4 [−6.1, 35.3]	−0.49 to +0.48
Clustered misses	2.46 to 25.15	3.35 [3.18, 3.54]	2.91	−9.1 [−50.0, 26.5]	−0.30 to +0.30
Imperfect clearing	2.83 to 30.07	3.41 [3.24, 3.58]	3.35	13.3 [−29.4, 51.6]	−0.49 to +0.43

simulation was run at the interpolated probability.

Estimator. With pooled control and policy rates B_ℓ, B_u and P_ℓ, P_u at p_ℓ, p_u, and each log rate assumed locally linear in $\log p$,

$$w = \frac{\log 3 - \log B_\ell}{\log B_u - \log B_\ell}, \qquad p^* = p_\ell^{1-w} p_u^w, \qquad F_{\mathrm{cb}} = 1 - \frac{P_\ell^{1-w} P_u^w}{3}. \tag{T11}$$

The interpolated baseline equals 3 by construction; this standardization does not test whether the simulator attains that rate at p^*. Whole seeds were resampled within banks A and B (128 each), 50,000 draws, seed 20261016; shared seed lists were resampled synchronously across conditions, and the independent and sensitivity lists independently. Rates, w and F_{cb} were recomputed from pooled counts and patient-days in each draw. The intervals cover Monte Carlo uncertainty conditional on the input probabilities and the interpolation rule; they exclude interpolation error, original probability selection, structural uncertainty and clinical-input uncertainty.

The three contrasts were 11.7 [4.0, 19.2] (independent minus clustered), 36.7 [26.8, 46.2] (independent minus imperfect) and 25.0 [15.3, 34.5] (clustered minus imperfect) percentage points; these are Bonferroni-adjusted 98.333% percentile bootstrap intervals within this secondary family of three, not tests of actual baseline equivalence. The target lay outside the resampled bracket, so the estimator extrapolated, in 2,267 (4.534%), 21 (0.042%) and 1,922 (3.844%) of 50,000 draws; extrapolating draws were retained and no draw was undefined.

Descriptive checks. Arithmetic instead of logarithmic interpolation shifts F_{cb} by at most 0.03 percentage points. Three-point curvature shifts the clustered and imperfect estimates by −0.21

Table T29: Post hoc common-baseline estimates (256 seed-matched pairs at each bracketing probability per profile). Target 3.0 is imposed by interpolation, not verified by new simulation. The absolute decrease equals $3F_{\mathrm{cb}}$, not an observed count difference at p^*. The original 512-pair count benefits remain 3.13, 2.46 and 1.24 acquisitions prevented per 56-day run at the original probabilities.

Profile	Baseline bracket	Implied p^*	Prevention (%) [95%]	Absolute decrease [95%]
Independent misses	2.777–7.289	0.145853	66.7 [62.1, 71.2]	2.00 [1.86, 2.14]
Clustered misses	2.630–5.029	0.079545	55.0 [50.7, 59.4]	1.65 [1.52, 1.78]
Imperfect clearing	2.809–5.928	0.076328	30.0 [23.6, 36.6]	0.90 [0.71, 1.10]

and −0.07 points (the independent profile has no third matched condition). Using the full 512-seed lower conditions gives 67.8% [64.8, 70.8], 55.5% [52.3, 58.7] and 27.5% [23.0, 32.0]; these intervals were corrected after review to preserve the overlap between the independent upper seeds and the lower bank (a dated amendment is archived; the primary analysis is unchanged). At a target of 2.7, 72.0% of independent and 84.1% of imperfect draws extrapolate, so that check cannot be presented as interpolation throughout the target band. A review-requested Linux-only subset of 192 independent pairs gave 66.6% [61.3, 72.0], with 18.8% of draws extrapolating; this changes the seed subset as well as the platform mixture and does not establish platform equivalence (Section T12). These checks do not bound interpolation or model error.

Verification. Independent implementations checked the points and intervals; the later raw-data reimplementation was not blind. Numerical agreement does not convert any failed matching check into a pass. Detailed comparison tolerances and record checks are in the execution records.

T7 Alternative-workflow study

T7.1 Protocol

The protocol preceded this study's outcomes, with original-workflow results and limitations already known; it is prospective for its own outputs, not for the earlier results. It asked whether the disagreement among the three hygiene profiles persists when every bedside site is cleared as its patient leaves (discharge or relocation) and detected carriers move only into empty isolation beds (Table T3). If no isolation bed is empty the carrier stays in its shared bed with that bed's multiplier; there are no occupant swaps and no added contact-precaution effect. Site clearing removes contamination and its origin label without resetting the daily cleaning clock; initial admission to a never-occupied bed adds no cleaning event. All other source, admission, staffing, transfer, hygiene and timing rules are inherited. The rules apply in warmup and observation. This is one combined alternative world: the design cannot attribute any change to vacancy cleaning or to the isolation rule separately, and original-versus-alternative results are descriptive structural sensitivity, not an isolated causal effect.

Fixed workload and selection. The baseline grid used 0.5, 1, 2 and 4 times each profile's original probability with 128 shared fresh seeds (1,536 baseline records; 90 warmup and 56 observation days at completion .90). Each probability was selected once from pooled baseline acquisitions per 1,000 patient-days, target 3: the lowest-p exact hit, otherwise the lowest-p adjacent increasing bracket with both rates positive, interpolating log rate linearly in $\log p$. No extrapolation, pseudocount,

alternative interpolator, fallback, grid expansion or use of policy outcomes was allowed. After grid validation the selection and derived policy configuration were hash-bound to the freeze and to every baseline record. Each profile then received 512 fresh paired control/policy seeds, shared across profiles and disjoint from the grid seeds; state was cloned after warmup and completion was .90 versus .95 with memory retained. The scientific ceiling was 3,072 records and was used exactly.

Inference. Pooled rates, F, absolute rate change and mean acquisitions prevented; 20,000 synchronous whole-seed bootstrap draws (seed 20261017; grid and policy streams from NumPy seed sequences with entropy [20261017, 0] and [20261017, 1]); 98.333% intervals for the fixed three-contrast Bonferroni family and descriptive 95% intervals otherwise. Baseline 95% intervals were checked against the band [2.7, 3.3] as a descriptive target-band check, not a profile-equivalence test, with no refitting or added seeds allowed. Policy intervals condition on the once-selected p; they do not propagate selection uncertainty or interpolation-model error.

T7.2 Selection

Table T30: Alternative-workflow baseline grid (128 seeds per point): acquisitions per 1,000 patient-days (acquisition counts in parentheses) at $m\times$ the original probability, and the once-selected probability. Every profile used the lowest positive increasing adjacent bracket (m1–m2) with log–log interpolation; there were no exact hits, nonincreasing segments or zero lower rates. The selection bootstrap (20,000 draws) had no failed or undefined draws; the last column counts draws that selected the lower (m0.5–m1) bracket.

Profile	m0.5	m1	m2	m4	Selected p (×m1)	Conditional 95%	Lower-bracket draws
Independent	0.205 (43)	1.919 (403)	17.03 (3,574)	96.73 (20,303)	0.165111 (1.152)	[0.15999, 0.17033]	0
Clustered misses	0.433 (91)	2.463 (517)	25.15 (5,280)	161.72 (33,952)	0.080633 (1.061)	[0.07801, 0.08332]	7
Imperfect clearing	0.538 (113)	2.828 (594)	30.07 (6,312)	168.58 (35,387)	0.076144 (1.017)	[0.07338, 0.07862]	2,861

Imperfect clearing chose the lower bracket in 14.3% of selection draws, yet its conditional p interval is still narrow (about ±3.5%). The selected p values are defined to within 1 to 4 units in the last place, depending on the summation order of patient-days; all 1,536 policy records use the cluster's p exactly.

T7.3 Results

Table T31: Holdout results in the two workflows (512 paired seeds per profile and workflow). Rates per 1,000 patient-days in the 56-day observation window; acquisitions are pooled control→policy counts; mean prevented is acquisitions prevented per 56-day ICU run. Descriptive 95% intervals; original-workflow intervals use 200,000 whole-seed draws, alternative-workflow intervals 20,000. The two studies differ in rules, selected p, seeds and execution; the side-by-side display is context only and defines no between-workflow estimand.

Profile	Workflow	p	Control rate	Policy rate	Acquisitions	F (%)	Rate reduction	Mean prevented
Independent misses	Original	0.14328	2.83 [2.66, 3.01]	0.92 [0.83, 1.00]	2,375 → 770	67.6 [64.1, 70.8]	1.91 [1.73, 2.10]	3.13 [2.84, 3.44]
	Alternative	0.16511	2.90 [2.76, 3.05]	0.68 [0.62, 0.75]	2,438 → 573	76.5 [74.1, 78.7]	2.22 [2.07, 2.37]	3.64 [3.39, 3.89]
Clustered misses	Original	0.07602	2.69 [2.54, 2.83]	1.19 [1.10, 1.28]	2,255 → 995	55.9 [52.1, 59.5]	1.50 [1.35, 1.66]	2.46 [2.21, 2.72]
	Alternative	0.08063	2.62 [2.48, 2.77]	1.21 [1.12, 1.31]	2,201 → 1,019	53.7 [49.7, 57.5]	1.41 [1.25, 1.56]	2.31 [2.06, 2.56]
Imperfect clearing	Original	0.07484	2.80 [2.66, 2.95]	2.05 [1.92, 2.17]	2,352 → 1,718	27.0 [21.6, 32.1]	0.76 [0.58, 0.93]	1.24 [0.96, 1.52]
	Alternative	0.07614	2.63 [2.48, 2.77]	1.91 [1.80, 2.03]	2,205 → 1,604	27.3 [22.2, 31.9]	0.72 [0.56, 0.87]	1.17 [0.93, 1.42]

Table T32: Prespecified between-profile contrasts in F (percentage points) with 98.333% intervals (fixed three-contrast Bonferroni families, one per study). Contrasts compare fixed configurations whose baselines differ; they are not contrasts at a common baseline. No original-versus-alternative difference is an estimand.

Contrast	Original workflow [98.333%]	Alternative workflow [98.333%]
Independent − clustered	11.7 [5.5, 18.0]	22.8 [17.1, 28.5]
Independent − imperfect	40.6 [33.1, 48.4]	49.2 [42.7, 56.0]
Clustered − imperfect	28.9 [21.1, 37.0]	26.4 [18.9, 34.3]

All three alternative-workflow contrasts exclude zero at 98.333%, ordering independent > clustered > imperfect (Tables T31 and T32); no bootstrap statistic was undefined. Only the independent baseline passed the descriptive target-band check. Clustered and imperfect baselines were below 2.7 and happened to differ by less than .005; they were not matched. Results condition on selected p.

The 128-seed m1 grid estimates were imprecise and overdispersed (variance/mean about 1.9): 2.46 [2.18,2.77] for clustering and 2.83 [2.51,3.15] for imperfect clearing. The latter selected p was only 1.7% above m1, but its holdout baseline was 2.63, suggesting an upward grid fluctuation. For clustering, grid error cannot be separated from curvature (log–log slopes 2.5 below and 3.35 above m1). Neither source enters policy intervals; Section T5.4 estimates the implied bias under stated assumptions.

Workflow counters. Mean per 56-day record (control / policy): isolation relocations (equal to relocation clears) 6.74 / 5.29, 6.62 / 5.80 and 6.70 / 6.21 in the three profiles; detected-patient-days spent waiting in a shared bed because no isolation bed was empty 1.68 / 1.25, 1.68 / 1.42 and 1.81 / 1.58 (per-tick counter divided by 2,592 ticks per day; it counts patient-ticks, not patients); discharge clears that removed contamination 5.90 / 4.31, 2.31 / 1.75 and 2.22 / 1.88; relocation clears that removed contamination 2.52 / 1.63, 1.08 / 0.70 and 1.03 / 0.81; contaminated vacancy clears were 6.2–8.1% of depositions. Discharge clears equalled discharges (about 311 per record), no occupied destination was ever attempted, and scheduled clears were 1,680 in every record. The waiting time, about 1% of colonized patient-days and 0.1% of all patient-days, bounds how often the background differs from the original workflow, not how much K differs; Section T4.3 reports the directly computed coefficient.

Face-validity and resources (post hoc). Table T33 gives alternative control quantities. Site contamination differs about threefold at similar acquisition baselines, suggesting positivity could constrain p separately from the hygiene law. A reported 2.1% environmental positivity (Wei et al., 2021) offers face validity only: organisms, sampling and binary sites differ. About 17,000 extra completions per run give threefold different returns per acquisition prevented. Natural contamination episodes recover stationary q_1, q_2 here because pickups are independent of completion state and contamination does not change behaviour. Episode-selection bias remains a real-world concern.

T7.4 Verification

Production selection and analysis used the pinned Linux runtime. Post-collection checks are summarized in Table T34; exact receipts and comparison tolerances are in the repository execution records. These are internal automated and delegated checks, not external peer review.

Table T33: Descriptive face-validity and resource quantities, alternative-workflow control arms (512 records per profile, 56-day observation window; descriptive, no intervals). Natural-episode survival: survived over at-risk counts at the first and second opportunity after pickup, stationary values in parentheses.

Quantity	Independent	Clustered	Imperfect
Colonized share of patient-days (%)	9.5	9.2	9.4
Mean fraction of bedside sites contaminated (%)	3.7	1.3	1.2
Mean census (occupancy %)	29.3 (97.6)	29.3 (97.6)	29.3 (97.6)
Hygiene opportunities per occupied patient-day	207.5	207.5	207.5
Care-zone visits per occupied patient-day	103.6	103.6	103.6
Patient contacts per occupied patient-day	101.1	101.1	101.1
Site contacts per occupied patient-day	207.1	207.1	207.1
Opportunities met with a contaminated hand (%)	0.94	0.36	0.39
Hand pickups from bedside sites (%)	42.6	21.3	20.0
Extra completed actions per run (policy − control)	17,022	17,015	17,018
Acquisitions prevented per run	3.64	2.31	1.17
Extra completed actions per acquisition prevented	4,700	7,400	14,500
Natural-episode q_1 (stationary)	0.1001 (0.1000)	0.1000 (0.1000)	0.2808 (0.2800)
Natural-episode q_2 (stationary)	0.0101 (0.0100)	0.0547 (0.0550)	0.0785 (0.0784)

Table T34: Post-collection checks of the alternative workflow.

Check	Finding
Inputs and records	Transport and frozen-input hashes agreed; all 1,536 grid records and 1,536 policy pairs passed completeness, counter and route checks.
Selection and analysis	Cross-platform frozen-code recomputation was bitwise; independent estimates agreed to numerical precision.
Runtime limitation	The unmodified frozen analyzer refused the local runtime. Cross-platform checks evaluated, but did not enforce, its pins; no second exact-runtime analysis followed collection.
Simulation recomputation	Five of 3,072 production records were re-simulated on macOS and agreed after runtime fields were removed.

T7.5 Execution limitations

A pre-production executable-identity failure produced no scientific records. A documented amendment changed runtime and provenance checks, leaving scientific functions unchanged, before the first record. All production records then used one Linux compute host and pinned package set. The amendment review was not blind. Seed-disjointness checks covered locally archived prior studies only; banks held only on the cluster were not checked. Absence of restarts was inferred from saved files because terminal scheduler accounting was not collected. Execution details are retained in the repository.

T7.6 Attribution limits

No original-versus-alternative difference has a declared estimand or interval. This includes the independent F (67.6% versus 76.5%), the independent-minus-clustered contrast (11.7 versus 22.8 percentage points), the selected probabilities (15%, 6% and 2% higher), the lower baseline at the original probability (independent grid m1 1.92 [1.70, 2.15] against the original holdout's 2.83 [2.66, 3.01]) and the near-disappearance of carry-over labels (Section T8). Each mixes two rule changes, a different selected p, fresh seeds and a separate study, so none is described as having grown, shrunk or been removed. The two rules are also entangled: with swaps a relocation never leaves a bed vacant, so separating them factorially would require an extra rule such as cleaning at swap. What follows from the model's rules rather than from estimated effects is that vacancy cleaning removes pre-admission contamination except deposits made while a bed is empty, that vacancy cleaning cannot change the $p = 0$ coefficient, and that the isolation rule can. The alternative workflow is one illustrative combination of rules, not a clinical practice and not a family of workflows.

T8 Route labels in both workflows

Labels. Passive histograms record, for every acquired colonization, the pickup that began the infecting hand's current uncleared contamination episode and the number of hygiene opportunities between that pickup and the transfer. Individual event traces were not saved. Four labels are used: *patient pickup*; *own-zone site, current stay* (pickup from the patient's own bedside site, contaminated during the current stay), split into 0 and more than 0 intervening opportunities; *other-zone site*; and *pre-admission carry-over* (pickup from the own bedside site where the contamination was deposited before this patient's admission). Counts are pooled over 512 paired seeds per profile and arm: the original-workflow counts come from the original holdout records and the alternative-workflow counts were re-tallied from the collected policy bank. Reduction is 1−policy/control count with a paired whole-seed 95% interval (20,000 draws, seed 20261021); patient-days differ by less than 0.010% between arms, so count and rate reductions agree to within 0.01 percentage points. Contribution is (control rate − policy rate of the label)/total control rate; it sums to F over the four labels.

Table T35: Route labels, original workflow (fixed holdout configurations). Control and policy counts with shares (%) of all acquisitions in that arm. †Fewer than 30 control acquisitions: the reduction is not interpretable and intervals at 0 or 100 are degenerate. ‡Some bootstrap draws were undefined (zero resampled control count): 53 of 20,000 for the clustered own-zone > 0 cell. The clustered carry-over cell further splits into 74 control acquisitions with 0 opportunities and 1 with more than 0; the latter's reduction is undefined in 7,417 of 20,000 draws and is not shown.

Profile	Label	Control n (%)	Policy n (%)	Reduction (%) [95%]	Contribution (pp)
Independent misses	Patient pickup	658 (27.7)	143 (18.6)	78.3 [74.1, 82.0]	21.7
	Own-zone site, current stay, 0 opp.	923 (38.9)	228 (29.6)	75.3 [69.9, 80.0]	29.3
	Own-zone site, current stay, > 0 opp.	0 (0.0)	0 (0.0)	n/a	0.0
	Other-zone site	344 (14.5)	80 (10.4)	76.7 [70.7, 82.0]	11.1
	Pre-admission carry-over	450 (18.9)	319 (41.4)	29.1 [17.8, 39.0]	5.5
	All acquisitions	2,375 (100.0)	770 (100.0)	67.6 [64.0, 70.8]	67.6
	All except carry-over	1,925 (81.1)	451 (58.6)	76.6 [73.3, 79.6]	62.1
	Site pickup, 0 opp. (own 0 + carry-over 0)	1,373 (57.8)	547 (71.0)	60.2 [54.9, 64.9]	34.8
Clustered misses	Patient pickup	1,361 (60.4)	590 (59.3)	56.6 [51.9, 61.0]	34.2
	Own-zone site, current stay, 0 opp.	489 (21.7)	222 (22.3)	54.6 [46.6, 61.8]	11.8
	Own-zone site, current stay, > 0 opp.	6 (0.3)	0 (0.0)	100.0† [100.0, 100.0]‡	0.3
	Other-zone site	324 (14.4)	130 (13.1)	59.9 [50.9, 67.6]	8.6
	Pre-admission carry-over	75 (3.3)	53 (5.3)	29.3 [0.0, 51.1]	1.0
	All acquisitions	2,255 (100.0)	995 (100.0)	55.9 [52.1, 59.4]	55.9
	All except carry-over	2,180 (96.7)	942 (94.7)	56.8 [53.0, 60.3]	54.9
	Site pickup, 0 opp. (own 0 + carry-over 0)	563 (25.0)	275 (27.6)	51.2 [43.5, 58.1]	12.8
Imperfect clearing	Patient pickup	1,467 (62.4)	1,047 (60.9)	28.6 [22.6, 34.3]	17.9
	Own-zone site, current stay, 0 opp.	492 (20.9)	382 (22.2)	22.4 [10.0, 33.0]	4.7
	Own-zone site, current stay, > 0 opp.	0 (0.0)	0 (0.0)	n/a	0.0
	Other-zone site	331 (14.1)	231 (13.4)	30.2 [17.6, 41.4]	4.3
	Pre-admission carry-over	62 (2.6)	58 (3.4)	6.5 [−36.0, 35.4]	0.2
	All acquisitions	2,352 (100.0)	1,718 (100.0)	27.0 [21.5, 32.0]	27.0
	All except carry-over	2,290 (97.4)	1,660 (96.6)	27.5 [22.0, 32.6]	26.8
	Site pickup, 0 opp. (own 0 + carry-over 0)	554 (23.6)	440 (25.6)	20.6 [8.5, 30.8]	4.8

Table T36: Route labels, alternative workflow (collected policy bank; once-selected p). Symbols as in Table T35; ‡56 of 20,000 draws undefined for the clustered own-zone > 0 cell. The imperfect own-zone > 0 cell (0 control, 1 policy acquisition) contributes −0.05 percentage points, shown as 0.0.

Profile	Label	Control n (%)	Policy n (%)	Reduction (%) [95%]	Contribution (pp)
Independent misses	Patient pickup	824 (33.8)	195 (34.0)	76.3 [72.6, 79.7]	25.8
	Own-zone site, current stay, 0 opp.	952 (39.0)	213 (37.2)	77.6 [73.9, 81.0]	30.3
	Own-zone site, current stay, > 0 opp.	0 (0.0)	0 (0.0)	n/a	0.0
	Other-zone site	597 (24.5)	137 (23.9)	77.1 [72.9, 80.9]	18.9
	Pre-admission carry-over	65 (2.7)	28 (4.9)	56.9 [34.5, 73.6]	1.5

Profile	Label	Control n (%)	Policy n (%)	Reduction (%) [95%]	Contribution (pp)
	All acquisitions	2,438 (100.0)	573 (100.0)	76.5 [74.1, 78.7]	76.5
	All except carry-over	2,373 (97.3)	545 (95.1)	77.0 [74.7, 79.3]	75.0
	Site pickup, 0 opp. (own 0 + carry-over 0)	1,017 (41.7)	241 (42.1)	76.3 [72.6, 79.7]	31.8
Clustered misses	Patient pickup	1,342 (61.0)	640 (62.8)	52.3 [47.4, 56.7]	31.9
	Own-zone site, current stay, 0 opp.	460 (20.9)	229 (22.5)	50.2 [41.7, 57.8]	10.5
	Own-zone site, current stay, >0 opp.	6 (0.3)	1 (0.1)	83.3† [0.0, 100.0]‡	0.2
	Other-zone site	371 (16.9)	142 (13.9)	61.7 [53.6, 68.8]	10.4
	Pre-admission carry-over	22 (1.0)	7 (0.7)	68.2† [30.0, 90.0]	0.7
	All acquisitions	2,201 (100.0)	1,019 (100.0)	53.7 [49.7, 57.5]	53.7
	All except carry-over	2,179 (99.0)	1,012 (99.3)	53.6 [49.5, 57.3]	53.0
	Site pickup, 0 opp. (own 0 + carry-over 0)	482 (21.9)	236 (23.2)	51.0 [42.7, 58.3]	11.2
Imperfect clearing	Patient pickup	1,404 (63.7)	1,029 (64.2)	26.7 [20.3, 32.5]	17.0
	Own-zone site, current stay, 0 opp.	438 (19.9)	329 (20.5)	24.9 [13.0, 35.5]	4.9
	Own-zone site, current stay, >0 opp.	0 (0.0)	1 (0.1)	n/a	0.0
	Other-zone site	347 (15.7)	238 (14.8)	31.4 [18.7, 42.3]	4.9
	Pre-admission carry-over	16 (0.7)	7 (0.4)	56.2† [0.0, 86.7]	0.4
	All acquisitions	2,205 (100.0)	1,604 (100.0)	27.3 [22.2, 32.0]	27.3
	All except carry-over	2,189 (99.3)	1,597 (99.6)	27.0 [21.9, 31.8]	26.8
	Site pickup, 0 opp. (own 0 + carry-over 0)	454 (20.6)	336 (20.9)	26.0 [14.5, 36.2]	5.4

Route-label results (descriptive). Tables T35–T36 show different hygiene responses within environmental labels. Original independent carry-over fell only 29.1% [17.8,39.0], while patient, current-stay own-zone and other-zone counts fell about 75–78%. Its share therefore rose from 18.9% to 41.4%. Excluding carry-over gives a 76.6% count reduction versus 67.6% overall; this is bookkeeping, not a counterfactual, because acquisitions seed later ones. Alternative carry-over counts were much smaller. Own-zone shares were similar between workflows and own-zone prevention remained large.

Current-stay own-zone pickups almost always have zero intervening opportunities (only 0–6 exceptions per arm), yet their reductions are within 6.3 points of patient-pickup reductions in every cell, with overlapping intervals. Earlier hand transport can cross hygiene opportunities before depositing into that zone. Carry-over can instead follow an ungated chain, but donor identities were not saved, so this explanation is untested. Prior-occupant transmission is clinically recognized (Mitchell et al., 2015); these binary sites are not calibrated to it.

The older pooled "site pickup with no intervening opportunity" metric mixes own-zone and carry-over labels. Its independent share rose 57.8%→71.0% entirely because carry-over rose; it is not a hygiene-insensitive fraction. Earlier surface shares (72.3/39.6/37.6%) equal the three site labels combined.

Labels describe realized paths, not causal fractions. Swapped-in patients count as current stay, not admission carry-over. In the alternative workflow, carry-over must have been deposited while the bed was empty. Workflow differences mix rules, probabilities, seeds and studies; they have no declared cross-workflow estimand or interval and identify neither rule's effect. Independent carry-over may be larger because its selected p is .143 versus about .076: clean-site chains need four transfers versus two for hand bridges, suggesting relative weight proportional to p^2, about 3.5-fold

higher. This remains a hypothesis.

Source status and gates (post hoc). Table T37 separates imported and acquired patient sources (20,000 paired whole-seed draws, seed 20261022). Imported-source reductions have intervals containing F^* in all six cells. Acquired-source intervals contain $1-(1-F^*)^2$, consistent with two gated hand segments. Counts are small (only seven independent-profile policy acquisitions from acquired sources), and the analysis followed outcomes. Product survival across gates could raise prevention above the limit, while ungated segments could lower it; these data do not establish that mechanism.

For count-based F, the identity $F^* - F = \sum_\ell \text{share}_\ell(F^* - \text{reduction}_\ell)$ assigns +8.72 of the original independent 7.55-point gap to carry-over (Table T38). This is a descriptive decomposition, not causal attribution.

Table T37: Patient-pickup acquisitions by source status (control → policy counts; fall in % with paired whole-seed bootstrap 95% interval). Imported-source falls are compared with the limit F^*, acquired-source falls with $1-(1-F^*)^2$. Post hoc and descriptive.

		Imported source			Acquired source		
Workflow	Profile	Counts	Fall (%) [95%]	F^*	Counts	Fall (%) [95%]	$1-(1-F^*)^2$
Original	Independent misses	557 → 136	75.6 [70.9, 79.9]	75.13	101 → 7	93.1 [86.2, 98.1]	93.8
	Clustered misses	1,204 → 552	54.2 [49.3, 58.8]	53.65	157 → 38	75.8 [65.2, 84.4]	78.5
	Imperfect clearing	1,264 → 938	25.8 [19.5, 31.7]	27.99	203 → 109	46.3 [31.0, 58.9]	48.2
Alternative	Independent misses	719 → 188	73.9 [69.8, 77.6]	75.18	105 → 7	93.3 [87.8, 97.9]	93.8
	Clustered misses	1,201 → 603	49.8 [44.9, 54.4]	53.65	141 → 37	73.8 [61.8, 83.2]	78.5
	Imperfect clearing	1,246 → 947	24.0 [17.6, 30.0]	27.99	158 → 82	48.1 [30.3, 62.0]	48.1

Table T38: Bookkeeping decomposition of the gap between the limit (75.13%) and the count-based prevented fraction (67.58%), original workflow, independent misses: share of control acquisitions (%), reduction (%) and contribution share × (limit − reduction) in percentage points. Not a causal attribution.

Label	Share	Reduction	Contribution (pp)
Patient pickup	27.7	78.3	−0.87
Own site, current stay	38.9	75.3	−0.06
Other-zone site	14.5	76.7	−0.23
Carry-over	18.9	29.1	+8.72
All labels	100.0	67.6	+7.55

T9 Environmental and surface-disabled comparisons

Environmental routes enter the leading-order expansion only beyond p^2 when sites start clean (Section T3), yet over a 146-day horizon they can matter at the selected probabilities. Two original-workflow experiments disabled hand–site exchange (in warmup and observation) at fixed probabilities. Removing these exchanges also removes everything downstream of them, so neither experiment isolates one pickup pathway, and neither matches baseline burden between the full and disabled models. Other transmission models have likewise found that the relative importance of

hand hygiene and environmental interventions depends on the routes represented (Spicknall et al., 2010; Barnes et al., 2014; Kraay et al., 2018).

Environmental comparison. At the three earlier independent-profile probabilities $p = .08098$, .10797 and .13497, 512 paired seeds ran with and without hand–site transmission (3,072 paired jobs). The three full-minus-disabled contrasts in F form one family at 98.333% (Table T39). The full model prevented a smaller fraction in the middle and high comparisons, but more acquisitions in absolute terms at all three probabilities, because it has more transmission to prevent: a larger percentage prevented need not mean more patients protected. The low-probability F contrast narrowly missed the planned descriptive precision goal of a 10-point half-width; the sample was not extended.

Table T39: Environmental comparison, independent misses, original workflow (512 paired seeds per condition). Left: condition results (rates per 1,000 patient-days; descriptive 95% intervals). Right: full-minus-disabled contrasts with 98.333% intervals; absolute differences in rate prevented (per 1,000 patient-days) and in mean acquisitions prevented per 56-day run.

p	Model	Control rate [95%]	Policy rate [95%]	F (%) [95%]	ΔF (pp)	$\Delta(R_0 - R_1)$	Δ mean prevented
0.08098	Full	0.395 [0.348, 0.445]	0.127 [0.104, 0.152]	67.8 [60.5, 74.1]	−10.2 [−22.4, 2.4]	0.10 [0.03, 0.18]	0.17 [0.05, 0.29]
	Disabled	0.211 [0.177, 0.246]	0.046 [0.032, 0.062]	78.0 [69.9, 84.8]			
0.10797	Full	1.000 [0.917, 1.089]	0.361 [0.317, 0.406]	63.9 [58.6, 68.8]	−11.7 [−20.7, −2.7]	0.34 [0.22, 0.46]	0.55 [0.36, 0.75]
	Disabled	0.401 [0.356, 0.447]	0.098 [0.076, 0.120]	75.7 [69.7, 81.1]			
0.13497	Full	2.288 [2.135, 2.445]	0.835 [0.758, 0.913]	63.5 [59.4, 67.3]	−14.3 [−21.1, −7.5]	0.99 [0.79, 1.20]	1.63 [1.30, 1.97]
	Disabled	0.591 [0.537, 0.645]	0.131 [0.108, 0.155]	77.8 [73.4, 81.8]			

Log–log slopes. Equal-weight three-point regressions of log rate on log p at exactly these probabilities (50,000 synchronized draws) give slopes of 2.03 [1.67, 2.42] for the environment-disabled baseline rate and 4.33 [3.75, 5.10] for the full-minus-disabled rate difference (Table T40). These are consistent with hand-only and higher-order environmental transmission at the tested probabilities, but three points cannot establish a limiting power or identify an environmental route fraction, and the full-minus-disabled difference includes downstream feedback.

Table T40: Descriptive equal-weight three-point regressions of log rate versus log p (p = .0809794281, .1079725708, .1349657135). Any nonpositive or undefined outcome would make a draw undefined; none was.

Cohort	Seeds	Outcome	Log slope [95%]	Undefined draws
pooled	512	Environment disabled	2.03 [1.67, 2.42]	0
pooled	512	Full minus disabled	4.33 [3.75, 5.10]	0
A	256	Environment disabled	2.02 [1.53, 2.57]	0
A	256	Full minus disabled	4.32 [3.54, 5.46]	0
B	256	Environment disabled	2.03 [1.53, 2.60]	0
B	256	Full minus disabled	4.34 [3.53, 5.61]	0

Surface-disabled comparison. At the original probabilities, the same 512 holdout seeds gave a resolved independent prevention increase of +9.7 [2.9,16.4] points (99.167%), but unresolved clustered and imperfect contrasts (Table T47). Independent absolute benefit fell despite the higher fraction prevented (Table T45). Shared seeds do not imply identical histories: removing surface transmission changes later feedback.

Leading-order rates. The calculation $1000p^2\overline{K}_h/D^{(0)}$, with $D^{(0)} = 1{,}640.36$, is close to surface-disabled point rates and far below full rates (Table T41). This is neither an error bound nor an equivalence test.

Table T41: Measured and leading-order rates (per 1,000 patient-days) and prevention, original workflow. Leading values are deterministic point calculations from the reported mean coefficients; measured intervals are paired bootstrap 95% intervals. No inference from interval overlap is implied.

Condition	p	Measured R_0 [95%]	Leading R_0	Measured R_1 [95%]	Leading R_1	Measured F [95%]	Limit F^*
Holdout, independent	0.143	2.829 [2.655, 3.006]	0.695	0.917 [0.835, 1.002]	0.173	0.676 [0.641, 0.708]	0.751
Holdout, clustered	0.076	2.686 [2.540, 2.834]	1.441	1.185 [1.096, 1.277]	0.668	0.559 [0.521, 0.595]	0.536
Holdout, imperfect	0.075	2.801 [2.658, 2.947]	1.587	2.046 [1.925, 2.169]	1.143	0.270 [0.216, 0.321]	0.280
Full environment	0.081	0.395 [0.348, 0.445]	0.222	0.127 [0.104, 0.152]	0.055	0.678 [0.605, 0.741]	0.751
Environment disabled	0.081	0.211 [0.177, 0.246]	0.222	0.046 [0.032, 0.062]	0.055	0.780 [0.699, 0.848]	0.751
Full environment	0.108	1.000 [0.917, 1.089]	0.395	0.361 [0.317, 0.406]	0.098	0.639 [0.586, 0.688]	0.751
Environment disabled	0.108	0.401 [0.356, 0.447]	0.395	0.098 [0.076, 0.120]	0.098	0.757 [0.697, 0.811]	0.751
Full environment	0.135	2.288 [2.135, 2.445]	0.616	0.835 [0.758, 0.913]	0.153	0.635 [0.594, 0.673]	0.751
Environment disabled	0.135	0.591 [0.537, 0.645]	0.616	0.131 [0.108, 0.155]	0.153	0.778 [0.734, 0.818]	0.751

Two mechanisms could explain departures from the hand-survival calculation: environmental pickup after entry washing can reach a patient before exit washing, and preventing one acquisition can remove a source of later transmission. These effects need not add, and their signs can change with susceptible depletion, saturation and isolation. Environmental persistence is plausible but its size needs separate evidence: resistant *Klebsiella pneumoniae* was recovered from uncleaned PVC for 20 to more than 43 days after drying an inoculum of about 10^6 colony-forming units, yet only 76 of 3,699 ICU environmental samples (2.1%) were positive (Wei et al., 2021), and persistence in moist reservoirs such as sinks (De Geyter et al., 2017) gives no bedside decay half-life. The model's binary sites, absence of natural decay, complete daily cleaning and clean starting state are modelling choices.

T10 Opportunity-rule (clock) studies

The two rules. The main model gives one washing opportunity on entering a care zone (the patient's bed and bedside site) and one on leaving it. A worker can therefore wash on entry, touch a contaminated site and reach the patient before the next opportunity. The inherited simulator instead offered washing before and after each object contact, which supplies six opportunities between the patient contacts of Fig. T2. A common completion percentage therefore refers to different opportunity denominators under the two rules.

Analytic comparator. For an isolated stationary bridge, raising completion from 90% to 95% reduces survival through two opportunities by 75.0%, 52.3% and 26.5% in the three profiles, and through six opportunities by 98.4%, 60.4% and 60.3% (exact values 63/64, and 60.376% and 60.343%). With six opportunities imperfect clearing is nearly tied with clustered misses, so the opportunity rule changes not only the size of the response but the relative standing of the profiles. These are survival calculations for specified bridges, not ICU forecasts.

Coarse inherited-clock sweep. A baseline-only grid tested the inherited clock for the three profiles at $p = .15$, .30, .60 and 1 (128 fresh seeds each; Table T42). An approximate sampled match required the whole 99.5833% simultaneous interval inside 2.7–3.3; none of the 12 conditions

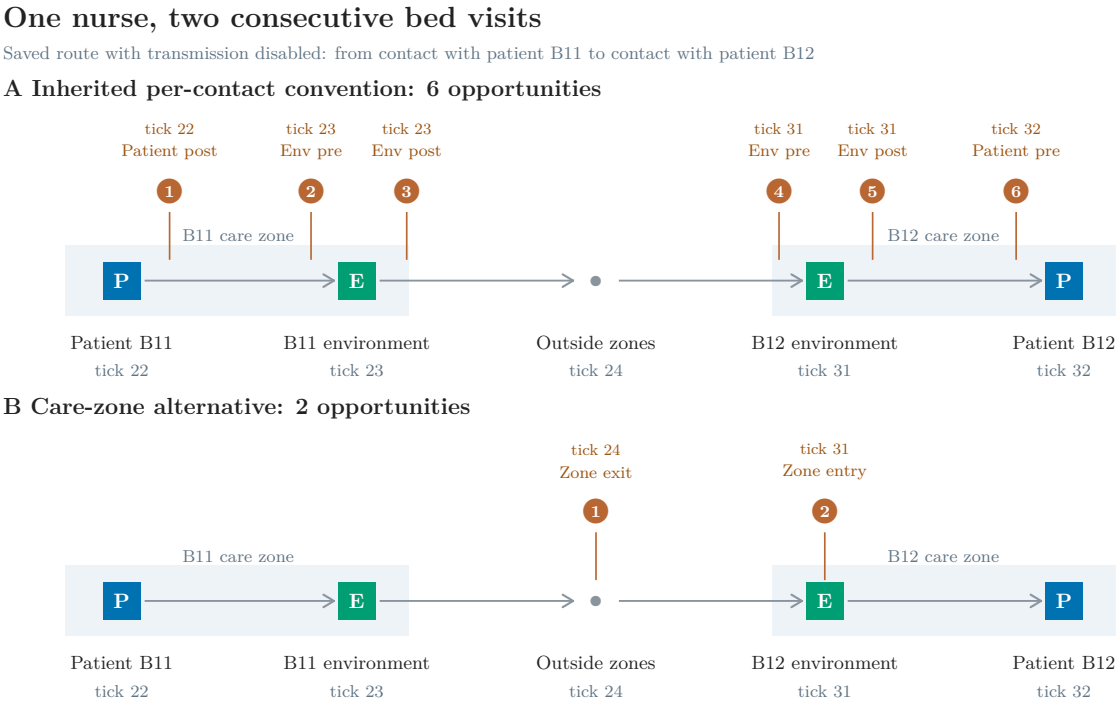


Fig. T2 A saved zero-transmission nurse route illustrates the opportunity conventions. The inherited object-contact rule gives six hygiene opportunities between two patient contacts; care-zone entry/exit gives two while retaining environmental contacts. This is a model trace, not a reconstructed clinical workflow.

met it. Independent completion stayed well below the target even at $p = 1$, while the clustered and imperfect rates jumped from near the target to far above it between adjacent grid points. This sampled grid does not prove a global maximum or that a match is impossible. Duplicated baseline fields in the clock analyzer do not estimate an intervention benefit and are not reported as policy outcomes. A condition with no observed acquisitions may have a degenerate zero interval; neither establishes a zero acquisition probability.

Table T42: Inherited-clock baseline sweep: 128 seeds per condition, rates per 1,000 patient-days with 99.5833% simultaneous intervals. No policy arm or refit.

Profile	p	Rate	Rate [99.5833% interval]	Sampled match
independent	0.15	0.00	0.00 [0.00, 0.00]	not established
independent	0.30	0.00	0.00 [0.00, 0.02]	not established
independent	0.60	0.06	0.06 [0.02, 0.12]	not established
independent	1.00	1.76	1.76 [1.42, 2.14]	not established
clustered misses	0.15	0.53	0.53 [0.39, 0.69]	not established
clustered misses	0.30	3.83	3.83 [3.28, 4.40]	not established
clustered misses	0.60	116.84	116.84 [113.92, 119.64]	not established
clustered misses	1.00	180.14	180.14 [177.42, 182.87]	not established
imperfect clearing	0.15	0.08	0.08 [0.03, 0.14]	not established
imperfect clearing	0.30	1.43	1.43 [1.15, 1.74]	not established
imperfect clearing	0.60	127.55	127.55 [124.22, 130.72]	not established
imperfect clearing	1.00	182.03	182.03 [179.51, 184.47]	not established

Opportunity-rule and matched-surface study. This separate study was frozen after the earlier results and research comments were known. Its 5,760 new records comprise a 19-condition inherited-clock baseline grid (128 shared seeds per condition; 2,432 records), two selected-clock comparisons (512 fresh pairs each), an unfitted independent $p = 1$ boundary (512 fresh pairs), three surface-disabled conditions at the original probabilities (512 matched old seeds each; Section T9) and an independent upper condition at 1.25 times the original probability (256 matched old seeds). Selection used the lowest exact hit, otherwise the lowest increasing adjacent bracket and linear interpolation, with no fallback, extrapolation, grid expansion or reselection; policy intervals condition on the once-selected p. The analysis used 50,000 whole-seed draws (seed 20261011); old vectors were resampled in four fixed 128-seed strata (bank A/B core and remainder; matched upper comparisons use the cores) and fresh clock vectors in independent A/B banks of 256. The six primary contrasts form one family at 99.167%; other intervals are descriptive 95%. A target check required the entire 99.167% baseline interval inside [2.7, 3.3]. New four-stratum intervals do not replace the archived original intervals elsewhere.

Table T43: Inherited-clock fine baseline grid (128 seeds per condition, A/B 64). Patient-days rounded for display. These point summaries select a probability; they do not certify target-band matching.

Profile	p	Acquisitions	Patient-days	Rate
Clustered misses	0.15	89	209,972	0.42
Clustered misses	0.169	136	209,925	0.65
Clustered misses	0.188	204	209,936	0.97
Clustered misses	0.206	233	209,965	1.11
Clustered misses	0.225	286	209,913	1.36
Clustered misses	0.244	435	209,928	2.07

Profile	p	Acquisitions	Patient-days	Rate
Clustered misses	0.262	532	209,948	2.53
Clustered misses	0.281	611	209,881	2.91
Clustered misses	0.3	821	209,908	3.91
Imperfect clearing	0.3	305	209,908	1.45
Imperfect clearing	0.312	387	209,883	1.84
Imperfect clearing	0.325	464	209,953	2.21
Imperfect clearing	0.338	583	209,938	2.78
Imperfect clearing	0.35	666	209,949	3.17
Imperfect clearing	0.375	1008	209,927	4.80
Imperfect clearing	0.4	1399	209,934	6.66
Imperfect clearing	0.45	3067	209,873	14.61
Imperfect clearing	0.5	5876	209,860	28.00
Imperfect clearing	0.6	26318	209,883	125.39

Table T44: Once-selected inherited-clock probability and selection uncertainty (A/B 64 resampling). Any failed draw makes the unconditional interval unavailable. No selection variability is propagated into policy outcomes.

Profile	Selected p	Conditional 95%	Failed draws	Unconditional 95%
Clustered misses	0.283	[0.274, 0.288]	1/50,000	unavailable
Imperfect clearing	0.345	[0.336, 0.352]	0/50,000	[0.336, 0.352]

Table T45: Follow-up condition results: control and policy rates per 1,000 patient-days and prevention (%), descriptive 95% intervals. Old full-512 points are unchanged; their four-stratum intervals are new diagnostics. The $p = 1$ condition is unfitted.

Condition	Control rate	Policy rate	F (%)
Care-zone: independent (old 512) $p = 0.143$, $n = 512$	2.83 [2.66, 3.01]	0.92 [0.83, 1.00]	67.6 [64.1, 70.8]
Care-zone: clustered misses (old 512) $p = 0.076$, $n = 512$	2.69 [2.54, 2.83]	1.19 [1.10, 1.28]	55.9 [52.1, 59.5]
Care-zone: imperfect clearing (old 512) $p = 0.0748$, $n = 512$	2.80 [2.66, 2.95]	2.05 [1.92, 2.17]	27.0 [21.6, 32.1]
Surface-disabled: independent $p = 0.143$, $n = 512$	0.70 [0.64, 0.76]	0.16 [0.13, 0.19]	77.3 [73.2, 81.0]
Surface-disabled: clustered misses $p = 0.076$, $n = 512$	1.41 [1.32, 1.50]	0.69 [0.63, 0.75]	51.4 [46.3, 55.9]
Surface-disabled: imperfect clearing $p = 0.0748$, $n = 512$	1.66 [1.56, 1.76]	1.20 [1.12, 1.28]	27.6 [21.4, 33.4]
Independent: upper (matched 256) $p = 0.179$, $n = 256$	7.29 [6.75, 7.85]	2.11 [1.90, 2.34]	71.0 [67.3, 74.4]
Independent: center (matched 256) $p = 0.143$, $n = 256$	2.78 [2.53, 3.04]	0.94 [0.82, 1.06]	66.3 [60.8, 71.3]
Inherited clock: clustered misses $p = 0.283$, $n = 512$	3.09 [2.91, 3.27]	0.86 [0.79, 0.94]	72.1 [69.4, 74.6]
Inherited clock: imperfect clearing $p = 0.345$, $n = 512$	2.85 [2.68, 3.03]	1.03 [0.94, 1.12]	64.0 [60.2, 67.5]
Inherited clock: independent boundary $p = 1$, $n = 512$	1.70 [1.57, 1.83]	0.029 [0.017, 0.042]	98.3 [97.5, 99.0]

Table T46: Absolute pooled rate reduction per 1,000 patient-days and mean paired count reduction per 56-day run, descriptive 95% intervals. Paired branches are not independent records.

Condition	n	Rate reduction	Mean count reduction
Care-zone: independent (old 512)	512	1.91 [1.73, 2.10]	3.13 [2.84, 3.44]
Care-zone: clustered misses (old 512)	512	1.50 [1.35, 1.66]	2.46 [2.21, 2.72]
Care-zone: imperfect clearing (old 512)	512	0.76 [0.59, 0.93]	1.24 [0.96, 1.52]
Surface-disabled: independent	512	0.54 [0.48, 0.60]	0.88 [0.79, 0.98]
Surface-disabled: clustered misses	512	0.73 [0.63, 0.82]	1.19 [1.03, 1.35]
Surface-disabled: imperfect clearing	512	0.46 [0.34, 0.58]	0.75 [0.56, 0.95]
Independent: upper (matched 256)	256	5.18 [4.61, 5.76]	8.49 [7.56, 9.44]
Independent: center (matched 256)	256	1.84 [1.57, 2.13]	3.02 [2.58, 3.48]
Inherited clock: clustered misses	512	2.23 [2.05, 2.41]	3.65 [3.36, 3.95]
Inherited clock: imperfect clearing	512	1.82 [1.63, 2.01]	2.99 [2.68, 3.30]
Inherited clock: independent boundary	512	1.67 [1.54, 1.80]	2.73 [2.52, 2.95]

Table T47: The fixed six-member primary family of the follow-up. Estimates and 99.167% intervals in percentage points of prevention; positive values mean greater relative prevention in the first configuration. An interval containing zero is unresolved, not equivalence or insensitivity.

Contrast	Estimate (pp)	99.167% interval
Inherited clock minus care-zone: clustered misses	16.3	[10.2, 22.3]
Inherited clock minus care-zone: imperfect clearing	37.0	[28.5, 45.7]
Surface-disabled minus full: independent	9.7	[2.9, 16.4]
Surface-disabled minus full: clustered misses	−4.5	[−13.1, 3.6]
Surface-disabled minus full: imperfect clearing	0.6	[−10.6, 11.5]
Upper minus matched center: independent	4.7	[−3.7, 13.7]

Results. Both selected-clock target checks failed (Table T20); the clustered descriptive 95% interval cannot replace the wider declared check. The resolved clock contrasts (Tables T45–T47) change both the rule and calibrated p, so do not isolate the clock's effect. The independent $p = 1$ boundary gives neither a global maximum nor proof of impossible fitting. In matched upper-versus-center independent runs, prevention rose 66.3%→71.0% as baseline rose 2.78→7.29; the 4.7-point contrast was unresolved. Its upward direction opposes a simple dilution explanation, whose mechanism remains unestablished.

Local p-sensitivity. The clustered and imperfect profiles were rerun at .75, 1 and 1.25 times their fixed probabilities with 256 fresh paired seeds per condition (Table T48). Varying p changed the baseline far more than the fraction prevented; none of the four low/high-versus-center contrasts was resolved, though the intervals allow changes of about 10 to 17 percentage points. The clustered point estimates decrease with p and the imperfect ones increase; neither trend is established, and a simple surface-dilution explanation does not account for the latter direction. The number of acquisitions prevented rose with the baseline. These records supply the elasticity slopes of Section T5.4.

Table T48: Local transmission-probability sensitivity, original workflow (256 fresh pairs per condition; 50,000 whole-seed draws within declared A/B banks, seed 20261010). Rates per 1,000 patient-days; mean prevented per 56-day run and absolute rate decrease with 95% intervals; low/high-minus-center contrasts in F with 98.75% intervals.

Condition	p	R_0	R_1	F (%) [95%]	Mean prevented [95%]	Rate decrease [95%]
Clustered, low	0.05702	1.30	0.51	60.5 [53.8, 66.4]	1.29 [1.07, 1.51]	0.78 [0.65, 0.92]
Clustered, center	0.07602	2.63	1.17	55.3 [49.9, 60.3]	2.39 [2.03, 2.76]	1.46 [1.24, 1.68]
Clustered, high	0.09503	5.03	2.33	53.6 [48.9, 57.9]	4.42 [3.86, 4.98]	2.69 [2.35, 3.04]
Imperfect, low	0.05613	1.25	0.89	28.9 [18.2, 38.2]	0.59 [0.35, 0.84]	0.36 [0.21, 0.51]
Imperfect, center	0.07484	2.81	1.97	29.8 [22.1, 36.8]	1.37 [0.97, 1.77]	0.84 [0.59, 1.08]
Imperfect, high	0.09355	5.93	3.99	32.7 [27.5, 37.5]	3.18 [2.57, 3.78]	1.94 [1.57, 2.30]
Contrasts [98.75%]: clustered low−center 5.13 [−4.91, 14.88]; high−center −1.76 [−10.72, 7.37]; imperfect low−center −0.88 [−17.58, 14.86]; high−center 2.91 [−8.09, 14.65].						

Opportunity intensity. The care-zone clock gives about 207 boundaries per occupied patient-day (8.6 per hour); merging direct exit/entry pairs gives about 104 WHO-style opportunities (4.3 per hour). Clinical rates use different units and tasks, including Moments 2 and 3; numerical similarity would not validate this rule. Table T49 gives model denominators. No full-ICU limit was computed for the inherited per-contact clock; finite-p clock comparisons also change selected p.

Table T49: Observation-period modeled counts by worker role (56-day window, pooled over the 256 coefficient backgrounds). The modeled opportunity intensity is not a clinical compliance denominator.

Role	Opportunities	Patient contacts	Surface contacts	Resets
Nurse	75780096	36931766	75679744	430080
Physician	8929028	4354559	8920247	157696
Ancillary (radiography)	2379776	1161740	2322432	114688

Role counts use observation events after tick 233,280, whereas event hashes cover all 146 days. They describe the declared model clock, not a validated clinical opportunity denominator.

T11 Measurement uncertainty study

Design and status. This synthetic assay study asks whether measuring q_2 as well as q_1 narrows policy forecasts when $h_0 = .90$ is known exactly. The protocol preceded assay draws, but analytic results were known; this is a method demonstration, separate from ICU simulations.

The candidate grid is $\rho = i/40$, $i = 0, \ldots, 40$, by $e = j/100$, $j = 0, \ldots, 100$: 4,141 stationary Markov laws, including the specified stationary mixture at $\rho = 1$ and $e = 0$ (all $q_k = 1$, $F = 0$). Four truths are $(0, 1), (.5, 1), (0, .8), (.5, .8)$. Separate assays initialize contamination independently of stationary completion; conditional clearing is independent. Equivalent independent Binomial(n, q_1) and Binomial(n, q_2) counts were drawn directly. Each truth used $n = 500, 2000, 8000$ measurements per assay and 1,000 repeated experiments, totaling 12,000, with no enlargement after outcomes. The PCG64 seed sequence was [2026101703, truth index, n]. Held-out targets were $h_1 = .925, .95, .975$ at $k = 2, 6$: $F = 1 - q_k(h_1)/q_k(h_0)$, keeping ρ, e fixed. No policy assay was used.

Confidence sets. Each exact two-sided Clopper–Pearson interval has at least 97.5% marginal coverage (Clopper and Pearson, 1934); their intersection therefore has at least 95% coverage without requiring assay independence. The q_1-only set retains grid points consistent with its interval; the nested joint set additionally requires q_2 consistency. Target bounds are grid minima and maxima, with membership tolerance 2×10^{-12} that can only widen sets. Inclusion of the true grid point covers all six targets simultaneously. Empty sets remain undefined and count as coverage failures; no substitute model is chosen.

Grid results. At 8,000 measurements per assay, adding q_2 narrowed the median $k = 2$ interval width (percentage points of prevention, .90→.95) from 25.00 to 5.10 (independent), 25.00 to 0.91 (clustered), 10.83 to 3.54 (imperfect) and 10.83 to 4.91 (combined), and the $k = 6$ widths from 48.44 to 2.39, 48.44 to 3.68, 16.21 to 5.84 and 16.21 to 7.57. All-six-target coverage was 95.2%, 96.1%, 96.2% and 97.3% (exact 95% Monte Carlo intervals [93.69, 96.44], [94.71, 97.21], [94.82, 97.30] and [96.10, 98.21]), with 26, 15, 9 and 0 empty joint sets per 1,000; there were 110 empty joint sets across all 12,000 experiments. Parameter-set coverage was 95.2%, 94.7%, 95.5% and 96.0%; a realized fraction below 95% is compatible with its sampling error. The second assay is not equally informative in every case: at $n = 500$ the combined mechanism's median $k = 2$ width did not decrease. Nesting guarantees that adding q_2 never widens a defined interval, not that it narrows it substantially. The joint experiment uses twice the assay budget of q_1 alone; this compares adding a different observable, not an equal-cost allocation.

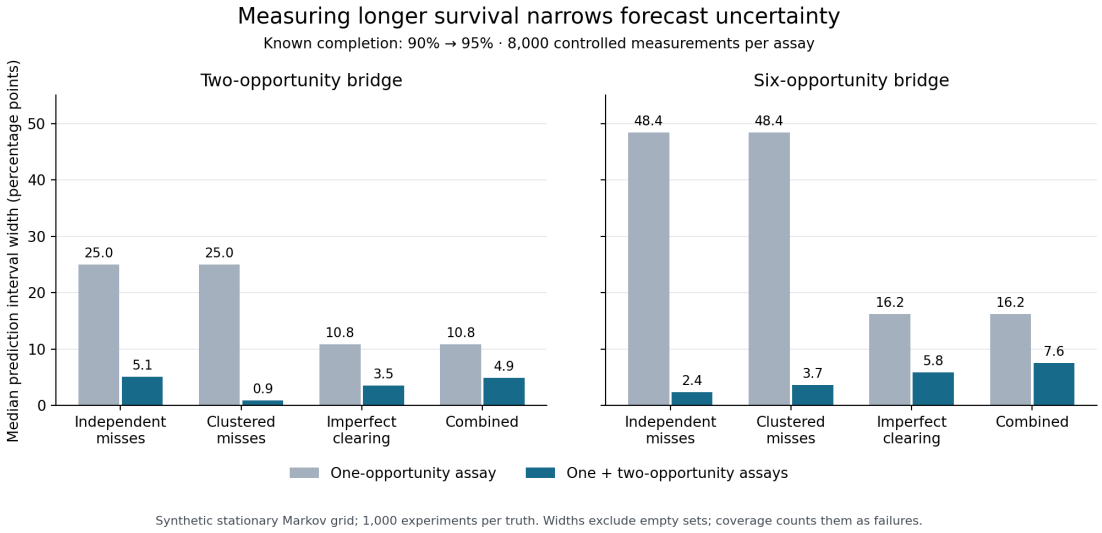


Fig. T3 Finite-grid measurement study: median projected interval widths for the held-out targets with q_1 only and with q_1 and q_2, by truth and sample size (1,000 repeated synthetic experiments per cell). Synthetic controlled assays with known h; conditional on the stationary first-order Markov family.

Continuous $k = 2$ bounds (post hoc). The grid understates widths. Reanalysis of saved .90→.95 measurements removed the grid restriction. For fixed e, the survival ratio is fractional-linear in ρ, so extremes occur at feasible endpoints. Assay constraints give one feasible e interval; prevention is nondecreasing in e along each clipped ρ boundary, so its extremes occur at that interval's ends. This argument holds for $0 < h_0 < h_1 < 1$ in this family.

Directed-rounding 60-digit arithmetic resolved all 24,000 confidence sets, with enclosures at most 10^{-58} wide; independent 100-digit calculations verified all 47,684 extrema and 158 empty-set decisions. This certifies the optimization given supplied limits, not the full statistical pipeline: Clopper–Pearson quantiles are floating point, widened by 3×10^{-12}. A verification-only 10^{-90} comparison tolerance repaired roundoff near a point certificate; the solver and outputs did not change.

Table T50: Continuous $k = 2$ bounds (.90→.95): median interval width (percentage points) with q_1 only and with q_1 and q_2, empty joint sets per 1,000 experiments, and single-target coverage at $n = 8{,}000$. Empty sets count as coverage failures. Coverage here concerns one $k = 2$ target and is not comparable with the six-target simultaneous coverage above.

Truth	Measurements per assay	q_1 only	$q_1 + q_2$	Empty joint sets	Coverage ($n = 8{,}000$)
Independent	500	25.71	19.28	15	
	2,000	25.16	11.50	16	
	8,000	25.04	6.27	22	95.2%
Clustered	500	25.71	5.68	14	
	2,000	25.16	2.45	10	
	8,000	25.03	1.13	15	97.0%
Imperfect	500	19.02	18.35	1	
	2,000	14.06	10.51	1	
	8,000	11.45	5.71	1	97.5%
Combined	500	19.01	18.97	0	
	2,000	14.07	12.98	0	
	8,000	11.45	6.69	0	99.8%

Table T50 gives continuous widths. Every defined grid interval lies within its continuous counterpart. There were 95 empty continuous joint sets versus 110 grid sets; all 15 newly feasible sets were retained. With perfect clearing, the roughly 25-point q_1-only width reflects the assumed $\rho \in [0, 1]$ support (F_2 ranges 75–50%), rather than data precision. Small samples give little gain for imperfect and combined truths. Six-opportunity results remain grid-conditional (Fig. T4).

Prespecified non-Markov stress test. The protocol also fixed a non-Markov alternative with the same h, q_1 and q_2 as the combined truth at every h, including the policy values. At each arm an episode-level completion propensity H equals $(1-\rho)h$ with probability $w = (1-h)/(1-h+\rho h)$ and 1 otherwise, with $\rho = .5$, $e = .8$, and completions independent given H. Then $\mathbb{E}[H] = h$ and $\mathrm{Var}(H) = \rho h(1-h)$, so q_1 and q_2 match the Markov model exactly, while $q_k = w(1-e(1-\rho)h)^k + (1-w)(1-e)^k$ generally differs for $k \geq 3$. Its $k = 2$ prevention is identical, but at $h = .95$ its $k = 6$ prevention is 56.42% against 58.73% in the matched Markov model. Reusing the combined truth's assay counts (their distribution is identical), the Markov-projected $k = 6$ intervals included the alternative truth in only 86.8% of experiments at $n = 8{,}000$ (Monte Carlo 95% interval [84.54, 88.84]); at smaller n the wider intervals covered it more often. There is no nominal coverage promise for this alternative, and the figure is not an estimated clinical misspecification rate. Better measurement can therefore produce narrower, confidently model-dependent predictions unless the dependence assumption is also assessed.

Projection code check. The widths above are for isolated bridges. To project saved assay limits onto an ICU limit surface, each experiment's feasible e follows from the q_1 interval and, at each e, the feasible ρ from the q_2 interval; extremes are taken over 81 values of e and 61 of ρ. Applied to the isolated bridge, this code reproduces the certified continuous widths (Table T51).

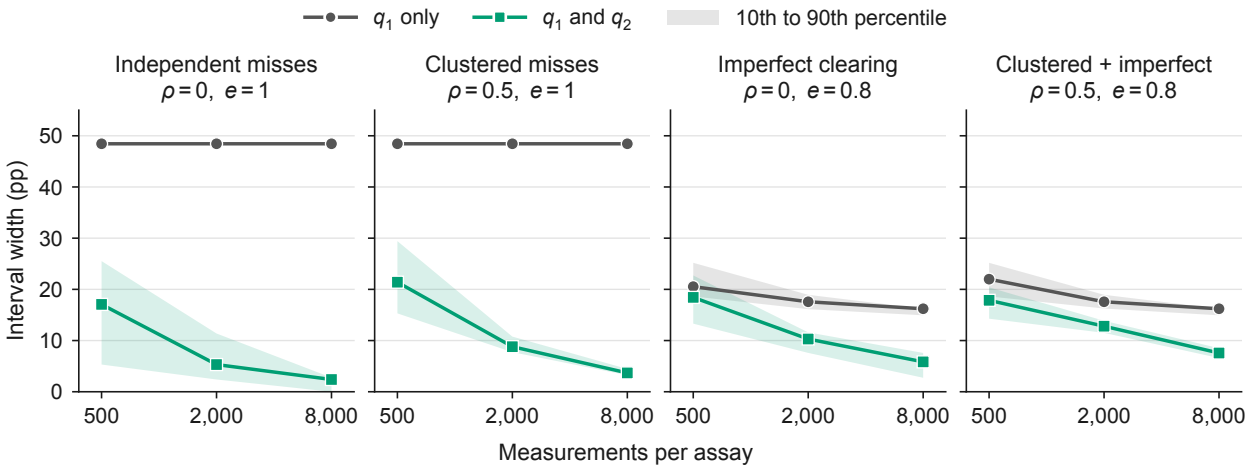


Fig. T4 Six-opportunity bridges: median width of the projected interval for prevention when completion rises from 90% to 95% ($h_1 = .950$ target), conditional on the finite 4,141-point grid, which can understate widths (it does so for $k = 2$). Grey: q_1 only; green: q_1 and q_2; bands, 10th to 90th percentiles over 1,000 experiments per point; empty joint sets (110 of 12,000) excluded from widths and counted as coverage failures.

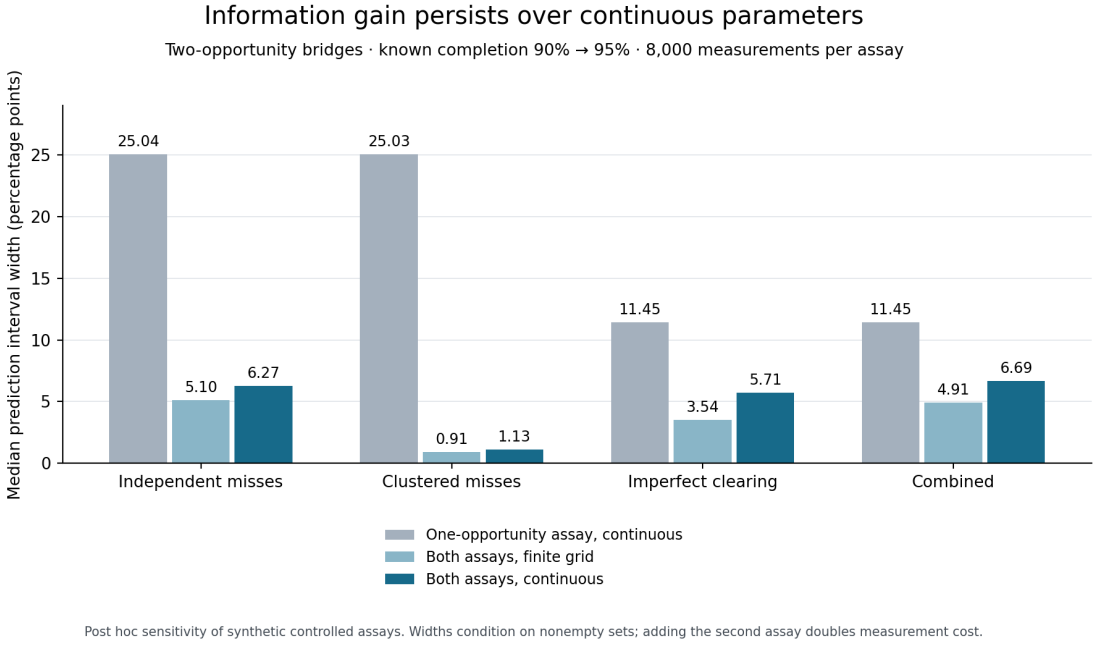


Fig. T5 Continuous versus finite-grid $k = 2$ interval widths for the same saved measurements (post hoc sensitivity; no new draws).

An earlier surrogate projection through a renewal-plus-residual surrogate surface is superseded by the prespecified projection through the computed coefficient grid below; the surrogate appears only in the protocol-specified comparison column of Table T52.

Table T51: Median interval widths (percentage points) at 8,000 observations per assay for the isolated two-opportunity bridge: certified continuous extrema, and the discretized projection code as a check. ICU projections are in Table T52.

Truth	q_1 only	$q_1 + q_2$	$q_1 + q_2$ (check)
Independent	25.0	6.27	6.27
Clustered	25.0	1.13	1.13
Imperfect	11.4	5.71	5.71
Combined	11.4	6.69	6.69

ICU grid projection (prespecified extension). The extension protocol reused the saved 8,000-per-assay limits, replacing the surrogate with bilinear interpolation of 132 computed ICU limits on $\rho \in [0, .95]$, $e \in [.5, 1]$. Search used 81 efficacy slices and 61 correlation points per nonempty slice. Sets outside the domain were excluded, without clipping or extrapolation; none occurred. An independent analytic check confirmed all 4,000 classifications. Empty and retained sets and conditional median widths appear in Table T52; the superseded surrogate is shown on those same sets only as the declared comparison.

The protocol's term "exact ICU projection" overstates the calculation. It interpolates sampled-history coefficients and discretizes feasible sets, without propagating background uncertainty or certifying continuous extrema. Ten separately computed crossing nodes gave post hoc discrepancies up to .656 points in F^*, not a global error bound or an interval-width bound. The isolated-bridge certificate and coverage guarantee do not transfer to this projection. No q_1-only ICU projection was computed, so these widths cannot establish ICU narrowing from adding q_2; the combined joint width 11.62 points is close to the isolated q_1-only width 11.4 points.

Table T52: Joint-assay ICU projection at 8,000 measurements per assay, 1,000 experiments per truth. Median widths (percentage points) are conditional on retained sets. The coefficient-grid column is interpolated and discretized, with no numerical error bound; the superseded surrogate column, on the same sets, is the protocol-specified comparison only.

Truth	Coefficient grid	Surrogate	Retained	Empty	Outside domain
Independent	5.26	6.17	978	22	0
Clustered	0.98	0.92	985	15	0
Imperfect	6.66	6.66	999	1	0
Combined	11.62	11.67	1000	0	0

Verification and limits. Independent code reproduced all random counts and defined bounds (285,924 endpoint comparisons; maximum discrepancy 2.45×10^{-15}); it followed results and was not blind. A separate summary check passed 834 assertions. These forecasts require known h, independently initialized assays, stationary first-order Markov completion, homogeneous clearing and unchanged ρ, e under policy. They omit measurement error in h and full-ICU finite-p, calibration and workflow uncertainty. Natural episodes can select completion states in practice, although they

do not in this model (Table T33). Clinical feasibility of these assays is unestablished. Individual completion streams plus technique or culture audits could instead estimate the parameters; lag-two autocorrelation distinguishes the Markov law (ρ^2) from the non-Markov mixture (ρ).

T12 Stability diagnostics and platform notes

These finite-window and bank diagnostics can reveal instability; flat-looking rates or intervals that include zero do not prove stationarity, sufficient warmup or equivalence. They are multiple unadjusted checks.

Observation halves. Tables T54 and T53 compare first and last 28-day rates. Original imperfect control rates fell (200,000-draw change interval $[-.64, -.08]$), while the inherited-clock independent $p = 1$ control rose. Other original control changes were unresolved. Alternative halves were computed post hoc from daily blocks (20,000 whole-seed draws, seed 20261023); its controls were stable-looking, but imperfect policy rates fell. These are finite-window diagnostics, not stationarity tests.

Table T53: Observation-half diagnostics (alternative workflow, post hoc): first-to-last 28-day rate per 1,000 patient-days and a descriptive 95% interval for last minus first.

Profile	Control	Change [95%]	Policy	Change [95%]
Independent misses	2.85 → 2.96	[−0.15, 0.37]	0.66 → 0.70	[−0.08, 0.15]
Clustered misses	2.62 → 2.63	[−0.27, 0.28]	1.21 → 1.22	[−0.15, 0.17]
Imperfect clearing	2.69 → 2.56	[−0.39, 0.14]	2.07 → 1.75	[−0.53, −0.12]

Table T54: Observation-half diagnostics (original workflow): first-to-last 28-day rate per 1,000 patient-days and a descriptive 95% interval for last minus first. All intervals have zero undefined draws.

Condition	Control first/last; change interval	Policy first/last; change interval
Care-zone: independent (old 512)	2.82 → 2.84 [−0.28, 0.33]	0.94 → 0.90 [−0.19, 0.12]
Care-zone: clustered misses (old 512)	2.73 → 2.64 [−0.37, 0.17]	1.18 → 1.19 [−0.15, 0.18]
Care-zone: imperfect clearing (old 512)	2.98 → 2.62 [−0.64, −0.084]	2.08 → 2.02 [−0.30, 0.18]
Surface-disabled: independent	0.72 → 0.67 [−0.17, 0.074]	0.17 → 0.15 [−0.081, 0.029]
Surface-disabled: clustered misses	1.45 → 1.37 [−0.25, 0.091]	0.72 → 0.66 [−0.18, 0.060]
Surface-disabled: imperfect clearing	1.70 → 1.62 [−0.29, 0.11]	1.19 → 1.22 [−0.13, 0.18]
Independent: upper (matched 256)	6.92 → 7.66 [−0.16, 1.65]	2.04 → 2.18 [−0.28, 0.56]
Independent: center (matched 256)	2.70 → 2.85 [−0.28, 0.58]	0.90 → 0.97 [−0.16, 0.30]
Inherited clock: clustered misses	3.04 → 3.14 [−0.23, 0.44]	0.87 → 0.85 [−0.17, 0.12]
Inherited clock: imperfect clearing	2.92 → 2.78 [−0.45, 0.19]	1.09 → 0.96 [−0.30, 0.040]

Condition	Control first/last; change interval	Policy first/last; change interval
Inherited clock: independent boundary	$1.58 \to 1.82$ $[0.010, 0.47]$	$0.026 \to 0.031$ $[-0.017, 0.029]$

Seed banks. Every study used two disjoint seed banks. In the original holdout, bank B minus bank A differences in F were -3.1 [-9.8, 3.6], -1.3 [-8.7, 6.0] and $+2.7$ [-7.9, 13.2] percentage points (independent, clustered, imperfect). In the environmental comparison one environment-disabled condition ($p = .081$) differed between banks by 17.0 [2.3, 33.0] points; the other five conditions' intervals included zero. Both banks were kept, as declared. Table T55 gives point diagnostics for the follow-up. In the coefficient studies the bank B minus bank A difference in every primary limit was at most 0.002 percentage points in either workflow.

Table T55: A/B bank point diagnostics of the opportunity-rule follow-up: control rates per 1,000 patient-days and prevention (%). Not formal between-bank equivalence tests. Matched upper/center banks contain 128 seeds each; full/fresh conditions 256 each.

Condition	A/B seeds	Control A/B	F A/B (%)
Care-zone: independent (old 512)	256/256	2.89/2.77	69.1/66.0
Care-zone: clustered misses (old 512)	256/256	2.72/2.65	56.5/55.2
Care-zone: imperfect clearing (old 512)	256/256	2.87/2.73	25.7/28.3
Surface-disabled: independent	256/256	0.67/0.72	74.5/79.9
Surface-disabled: clustered misses	256/256	1.44/1.39	48.2/54.6
Surface-disabled: imperfect clearing	256/256	1.56/1.76	22.9/31.7
Independent: upper (matched 256)	128/128	7.08/7.50	69.1/72.8
Independent: center (matched 256)	128/128	2.91/2.64	69.2/63.1
Inherited clock: clustered misses	256/256	3.07/3.11	73.1/71.1
Inherited clock: imperfect clearing	256/256	2.74/2.96	60.5/67.2
Inherited clock: independent boundary	256/256	1.67/1.72	98.1/98.5

Weekly control trajectories. The saved original-holdout trajectories were binned by week across the 90 warmup and 56 observation days, pooling counts and patient-days within each bin before division (Fig. T6, Table T56). Rates rose during the first four weeks of warmup and then varied between about 2.3 and 3.3 with no visible trend. Week 13 [84, 91) spans the day-90 boundary and the final bin [140, 146) has six days. Intervals are descriptive whole-seed bootstrap intervals (50,000 draws, seed 20261010). The bank-specific weekly tables are archived in the repository.

Table T56: Weekly baseline rates per 1,000 patient-days, original holdout, banks pooled.

Week	Days	Phase	Independent [95%]	Clustered misses [95%]	Imperfect clearing [95%]
1	[0,7)	warmup	1.22 [0.95, 1.50]	1.22 [0.98, 1.46]	1.11 [0.89, 1.35]
2	[7,14)	warmup	2.26 [1.84, 2.71]	1.55 [1.29, 1.83]	1.88 [1.55, 2.21]
3	[14,21)	warmup	2.31 [1.94, 2.71]	2.13 [1.79, 2.49]	2.50 [2.16, 2.84]
4	[21,28)	warmup	2.60 [2.21, 3.02]	2.51 [2.16, 2.87]	2.42 [2.09, 2.76]
5	[28,35)	warmup	2.73 [2.32, 3.16]	2.43 [2.09, 2.79]	2.87 [2.49, 3.26]
6	[35,42)	warmup	2.80 [2.38, 3.26]	2.80 [2.42, 3.19]	2.99 [2.59, 3.42]
7	[42,49)	warmup	2.53 [2.14, 2.94]	2.30 [1.98, 2.64]	2.86 [2.49, 3.24]
8	[49,56)	warmup	2.93 [2.52, 3.36]	2.50 [2.14, 2.87]	2.52 [2.18, 2.88]
9	[56,63)	warmup	2.93 [2.47, 3.45]	2.44 [2.09, 2.80]	3.15 [2.75, 3.56]
10	[63,70)	warmup	3.27 [2.81, 3.74]	2.38 [2.07, 2.71]	2.84 [2.48, 3.22]
11	[70,77)	warmup	2.55 [2.16, 2.97]	2.49 [2.14, 2.84]	2.90 [2.51, 3.31]

Week	Days	Phase	Independent [95%]	Clustered misses [95%]	Imperfect clearing [95%]
12	[77,84)	warmup	2.87 [2.45, 3.31]	2.85 [2.47, 3.25]	2.94 [2.57, 3.33]
13	[84,91)	mixed	2.74 [2.36, 3.14]	2.72 [2.36, 3.11]	2.80 [2.43, 3.18]
14	[91,98)	observation	2.57 [2.18, 2.98]	2.61 [2.26, 2.98]	3.06 [2.67, 3.46]
15	[98,105)	observation	3.06 [2.61, 3.54]	2.74 [2.39, 3.11]	3.15 [2.77, 3.55]
16	[105,112)	observation	2.92 [2.49, 3.37]	2.87 [2.50, 3.25]	2.87 [2.50, 3.26]
17	[112,119)	observation	2.65 [2.24, 3.09]	2.72 [2.36, 3.12]	2.90 [2.52, 3.30]
18	[119,126)	observation	2.72 [2.30, 3.16]	2.60 [2.23, 2.99]	2.48 [2.15, 2.82]
19	[126,133)	observation	3.12 [2.68, 3.57]	2.39 [2.05, 2.75]	2.70 [2.34, 3.07]
20	[133,140)	observation	2.67 [2.30, 3.06]	2.80 [2.44, 3.18]	2.67 [2.33, 3.02]
21	[140,146)	observation	2.92 [2.49, 3.38]	2.70 [2.30, 3.11]	2.57 [2.20, 2.96]

Execution platforms. The original holdout (1,344 Linux-cluster and 192 local macOS paired jobs) and the environmental comparison (2,688 and 384) mixed two platforms with identical scientific sources and package versions (Linux, Python 3.12.3; macOS arm64, Python 3.12.2; NumPy 1.26.4 and Mesa 1.2.1 on both). All 3,072 alternative-workflow records ran on one Linux host with one pinned build. The common-baseline analysis uses an independent lower bank of 64 macOS and 192 Linux records, with all upper and sensitivity records on Linux. Scientific source hashes and package versions agree, but exact cross-platform recomputation of those records was not verified, so platform effects could affect marginal estimates, not merely pairing efficiency; the intervals condition on the recorded execution mixture. A Linux-only subset of 192 pairs gave 66.6% [61.3, 72.0] against 66.7% in the primary analysis (Section T6); this changes the seed subset as well as the platform mixture. The alternative-workflow coefficient study ran on macOS (Section T13). The original-workflow coefficient backgrounds were produced on both platforms (38 partitions, 32 cluster and 6 local). The measurement study is a small local synthetic benchmark on macOS.

T13 Reproducibility

Archives. Frozen configurations, seed lists, source code, analysis scripts and records are preserved with checksums: the earlier 8,576 records, the 5,760 records of the opportunity-rule follow-up (whose analysis reuses the 1,536 original holdout records, not counted again), the 3,072 alternative-workflow records, the 256 alternative-workflow coefficient records and the 256 original-workflow coefficient-extension records. Paired branches are not separate records. The common-baseline analysis, the addenda and the measurement study add no ICU simulation records. Earlier exploratory studies remain archived, including failed fits and a 21-day calibration study whose fresh validation runs had lower baseline acquisitions than its training runs; an audit found no configuration or implementation difference, and no cause was identified. Its 21-day records enter no analysis reported here, and every later study used two disjoint, equally weighted seed banks (Section T12). Clinical spreadsheets from the original project are not public; every model input derived from them is encoded in the preserved configuration, and none is needed to reproduce any result.

Repository records and availability. The code, synthetic data and technical execution extracts are available at `https://github.com/ndandnd/icu-hygiene-calibration` (release v1.0.2), under `execution_records/`. Its `INDEX.md` maps the extracts to the original sections. It includes execution amendments, runtime details, replay checks, seed lists and summary hashes. This standalone report preserves the detailed scientific tables and execution record. The main article and its

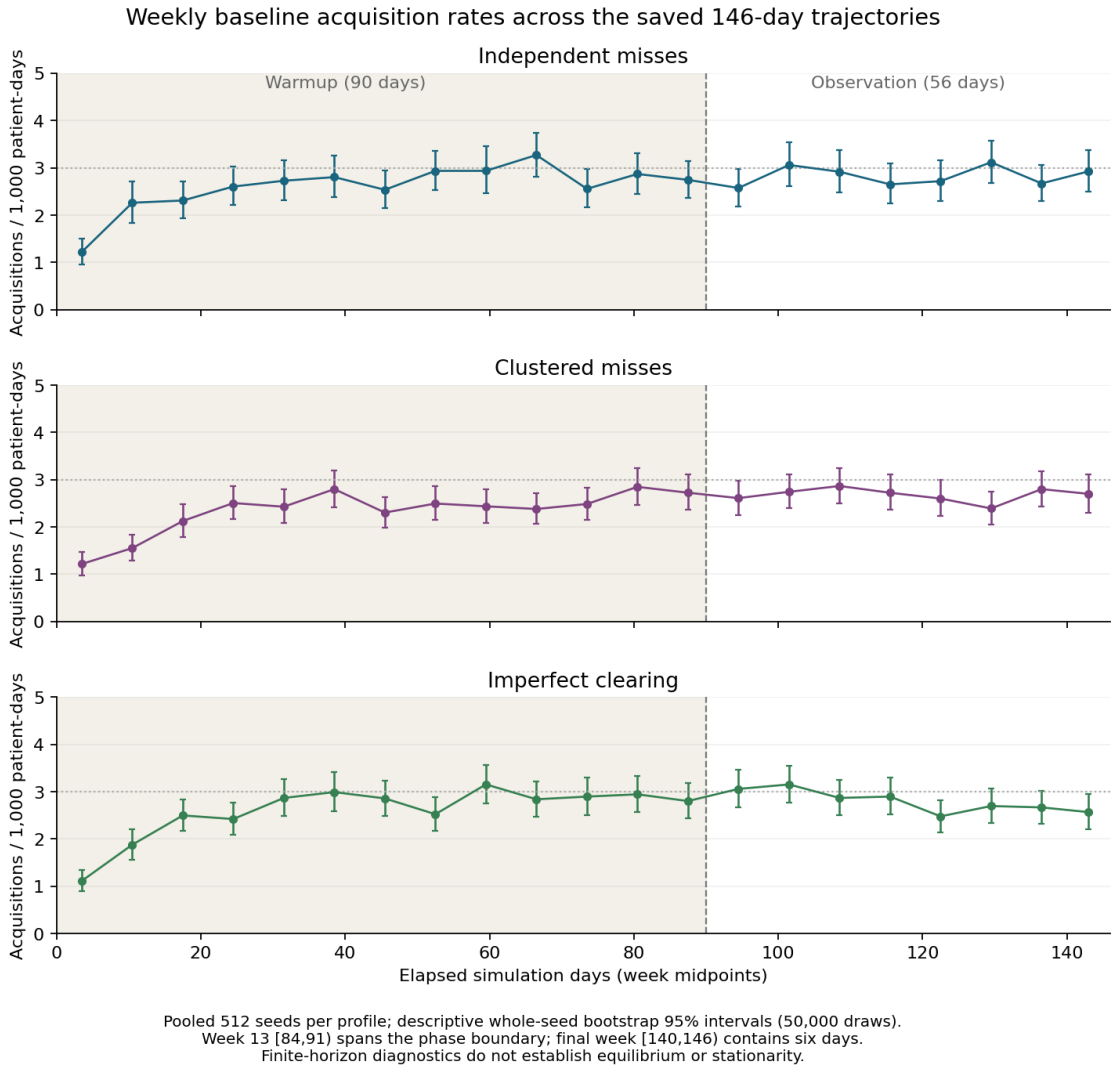


Fig. T6 Weekly control acquisition rates in the original holdout (512 seeds per profile; descriptive whole-seed bootstrap 95% intervals). Week 13 spans the day-90 boundary; the final bin has six days. The figure describes the saved finite horizon and does not establish equilibrium.

core supplementary PDF are distributed with the journal submission rather than this repository. Dataset descriptions and reproduction instructions are in the repository; no public DOI is claimed here.

Linux production used Python 3.12.3; local macOS runs used Python 3.12.2, with NumPy 1.26.4 and Mesa 1.2.1 on both. Cross-platform evidence and its limits are reported in Sections T7.4 and T12. The alternative-workflow coefficient study was local; original-mode recomputation covered five cluster-origin and five local-origin seeds, but alternative mode was not checked across platforms. The synthetic measurement study used macOS and SciPy 1.13.1.

Scientific tables remain generated from preserved inputs by `../supplement_numbers/make_supplement_tables.py` and `../supplement_numbers/make_extension_tables.py`. Their number-provenance maps bind displayed values to source fields, rounding and input/output hashes. Neither script was rerun for this edit.

T14 Directional transfer and bounded worker heterogeneity

This post hoc analysis uses the saved coefficient histories only. It adds no transmission simulation, background seed, bootstrap draw or clinical input. The stress ranges below were written down before reweighted outcomes were computed; the original aggregate outcomes were already known.

Global asymmetry and the limit. The common finite-p transfer parameter is a structural simplification. Its equality across directions is unnecessary for the leading hand-bridge ratio. Under the assumptions of Proposition T2, let patient-to-hand, hand-to-patient, hand-to-site and site-to-hand probabilities be respectively $\epsilon a, \epsilon b, \epsilon c, \epsilon d$, times the existing contact weights, with fixed bounded constants and $a, b > 0$. For sufficiently small ϵ these are valid probabilities. Exactly two successful transmissions can produce an acquisition only by imported patient–hand–susceptible patient transfer. Hence

$$\mathbb{E}C_h = \epsilon^2 ab\bar{K}_h + O(\epsilon^3), \qquad F \longrightarrow 1 - \bar{K}_{h_1}/\bar{K}_{h_0}.$$

Both coefficients acquire the same factor ab, which cancels. With initially clean sites, an environmental chain additionally needs hand-to-site deposition and site-to-hand pickup: its first possible contribution is fourth order when all four directions scale with ϵ. Cubic corrections need not be environmental. If surface exchange instead stays of order one, this is a different asymptotic regime. None of these identities validates directional robustness at the finite probabilities used in the simulations.

A conditional reweighting bound. Let bridge types have nonnegative baseline shares w_j summing to one and survival ratios $r_j = s_{1j}/s_{0j}$, with $s_{0j} > 0$. Suppose altered pickup and transfer weights multiply bridge j by $t_j \in [m, M]$, where $0 < m \le M$, identically in both hygiene arms. Then

$$F^*(t) = 1 - \frac{\sum_j w_j t_j r_j}{\sum_j w_j t_j}, \qquad |F^*(t) - F^*(1)| \le (r_{\max} - r_{\min})\frac{\sqrt{M} - \sqrt{m}}{\sqrt{M} + \sqrt{m}}.$$

This is a convenient corollary of bounded reweighting, not a new general probability theorem. To prove it, put $\mu = \sum_j w_j t_j$ and $u_j = w_j t_j/\mu$. A bounded mean changes by at most its range times

total variation. Convexity of $|t-\mu|$ on $[m, M]$ gives

$$\tfrac{1}{2}\sum_j |u_j - w_j| \le \frac{(M-\mu)(\mu-m)}{\mu(M-m)} \le \frac{\sqrt{M}-\sqrt{m}}{\sqrt{M}+\sqrt{m}},$$

with the last maximum at $\mu = \sqrt{mM}$; $m = M$ follows by continuity. Two types with endpoint multipliers and endpoint survival ratios attain the bound over unrestricted spectra. Attainability need not hold for a particular ICU or for factorized pickup/transfer weights. Types with zero baseline but positive policy coefficient fall outside this statement.

Saved-data calculation. Both workflows retain baseline and policy coefficients by worker in all 256 background records, for each of $(\rho, e) = (0,1), (.5,1), (0,.8), (.5,.8)$. For each worker u, pool these coefficients as A_u, B_u over backgrounds and compute $F^*(t) = 1 - \sum_u t_u B_u / \sum_u t_u A_u$. One multiplier applies to the same worker across seeds and both arms. A worker-specific pickup factor times its transfer factor can realize this product; varying their ratio alone cancels. No worker-role interpretation or within-worker contact heterogeneity is inferred.

We optimized the ratio over $t_u \in [1/2, 2]$ and $[1/4, 4]$, mathematical stress ranges without empirical or clinical status. Sort B_u/A_u and evaluate the endpoint assignments at each threshold to obtain the exact fractional-program extrema. An independent implementation solved the residual-root equations instead. All 790 bound input hashes, both complete 256-record banks, the original archived references and all worker sums passed verification. Exact rational arithmetic treats saved decimal coefficients as inputs; it does not certify the original coefficient engine's numerical error. The largest relative discrepancy between summed worker coefficients and recorded totals was below 4×10^{-13}.

Table T57: Conditional leading-order prevention (%) under worker reweighting. Endpoints are rounded outwards to two decimals. These are structural bounds on saved pooled coefficients, not confidence intervals. The combined profile has $(\rho, e) = (.5, .8)$.

Workflow	Profile	Nominal	$t_u \in [1/2, 2]$	$t_u \in [1/4, 4]$
Original	Independent misses	75.13	[75.04, 75.17]	[74.85, 75.19]
	Clustered misses	53.65	[53.60, 53.76]	[53.58, 53.89]
	Imperfect clearing	27.99	[27.97, 28.04]	[27.96, 28.09]
	Combined	37.01	[36.88, 37.32]	[36.83, 37.69]
Alternative	Independent misses	75.18	[75.18, 75.19]	[75.17, 75.19]
	Clustered misses	53.65	[53.60, 53.77]	[53.58, 53.89]
	Imperfect clearing	27.99	[27.96, 28.03]	[27.95, 28.08]
	Combined	37.00	[36.87, 37.32]	[36.80, 37.68]

All profile rankings retain their signs throughout either box in both workflows: independent misses, clustered misses, combined, then imperfect clearing. In particular, clustering lowers prevention with perfect clearing and raises it at $e = .8$. Subtracting separately optimized profile intervals gives conservative bounds for contrasts with the same worker multipliers; no jointly sharp contrast interval is claimed. The bounds condition on these backgrounds and omit their Monte Carlo uncertainty, finite-transmission feedback, arbitrary contact-specific heterogeneity and estimation of transfer parameters. They do not establish matched finite-p baselines or clinical validity.

The plan, mathematical derivation, exact check, input hashes, analysis and independent recomputation are preserved under `research/referee_perspective_2026-10-24/`. This calculation

changes none of the original studies, failed matching checks or coefficient predictions.

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